



Computer Networks

Introduction TCP/IP



Content

- Introduction to the TCP/IP
- Introduction to the IP Address
- TCP Three Way Handshake
- Mapping L2 Address to L3 Address



Overview of TCP/IP

- TCP/IP suite grew out of a 1969 DARPA funded project that look to develop a packet switching network – ARPANET
- 1983, ARPANET divided into two networks: MILNET and smaller ARPANET (Internet)
- 1990, ARPANET formally existence
- TCP/IP is available to many OS : Linux, MS-DOS, Windows, OS/2, Novell ,...



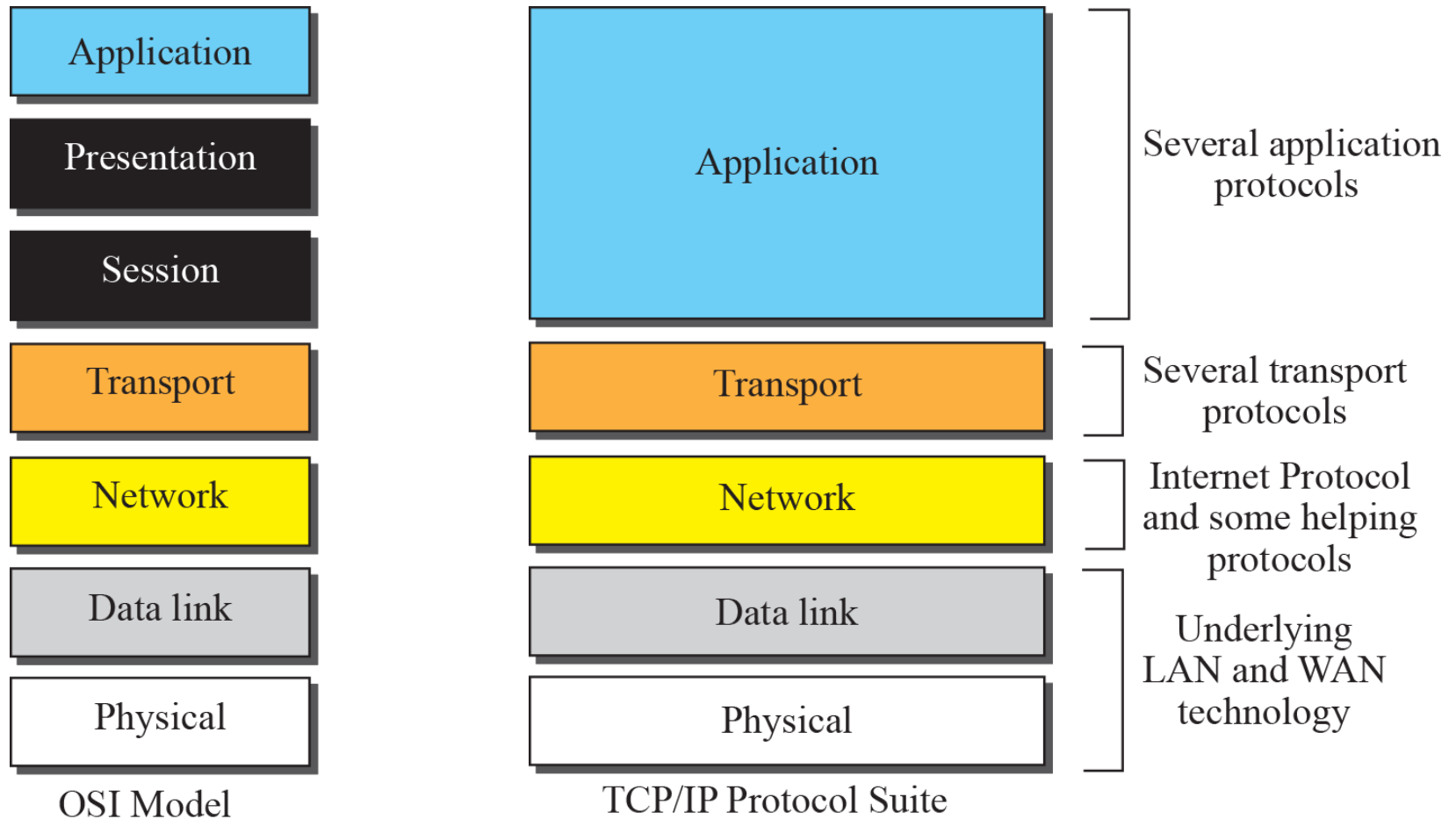
Overview of TCP/IP suite



- Protocols: IP (Internet Protocol), Transmission Control Protocol (TCP), User Datagram Protocol (UDP), Internet Control Message Protocol (ICMP), ...
- IP : defines the structure of packets of data that exchanged over the network
- TCP : byte stream protocol that support error checking, flow control and is extremely reliable. (Telnet, FTP, ...)
- UDP : unreliable and connectionless : do not “agree” to data being transmitted prior to the sender beginning transmission and no technique to verify arrived data. (NFS, NIS, DNS, ...)



OSI model and TCP/IP Suite





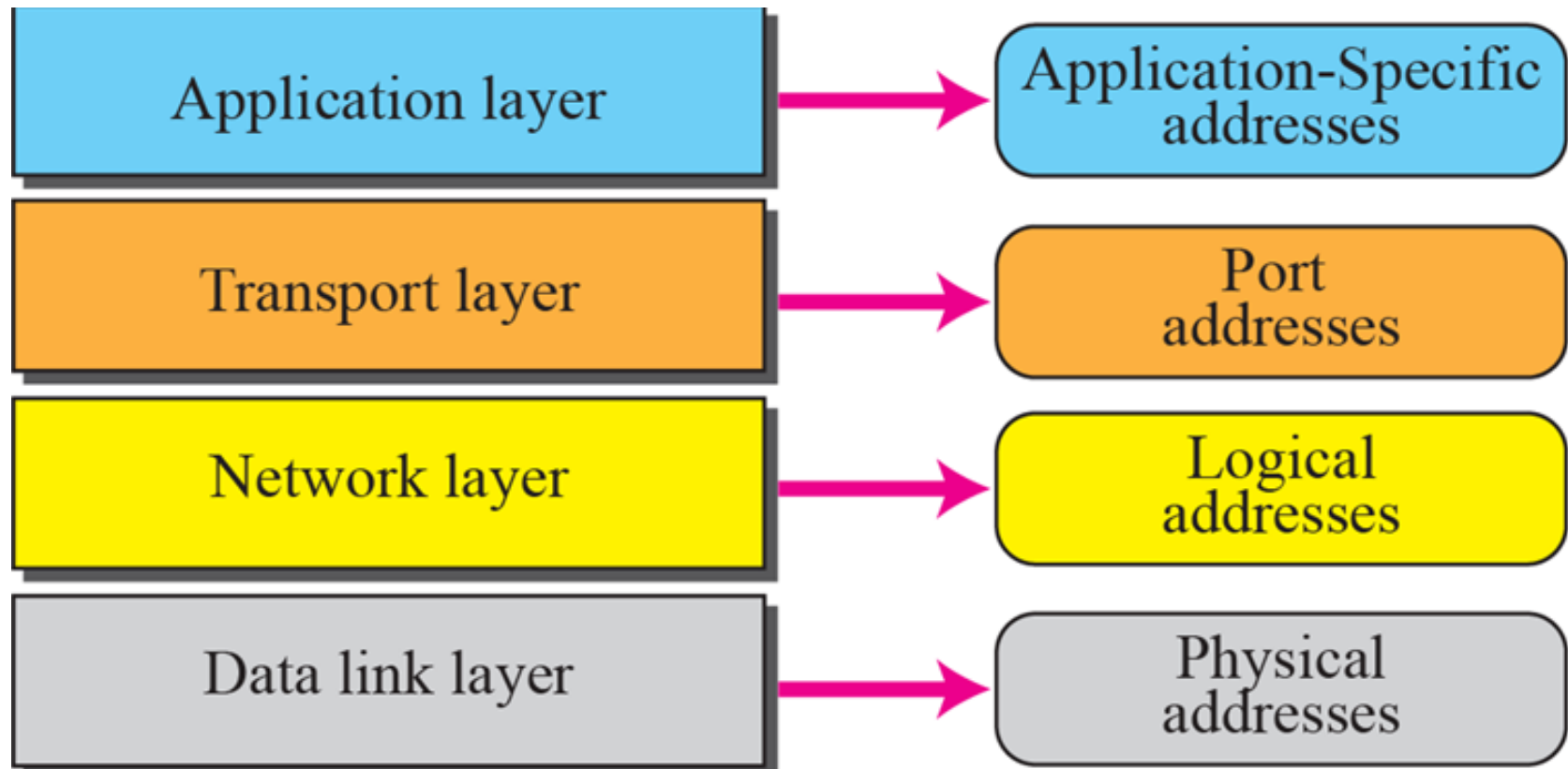
Addressing

Four levels of addresses are used in an internet employing the TCP/IP protocols suite:

- application-specific address
- port address (16 bits)
- logical address (IP address)
(IPv4 32bits, IPv6 128bits)
- physical address (48 bits - mac address)



Address in TCP/IP suite





Port Address

- The address given to each process (application) is called port address and it is an integer 16 bit number.



Port Address

-
- 21: [File Transfer Protocol](#) (FTP)
 - 22: [Secure Shell](#) (SSH)
 - 23: [Telnet](#) remote login service
 - 25: [Simple Mail Transfer Protocol](#) (SMTP)
 - 53: [Domain Name System](#) (DNS) service
 - 80: [Hypertext Transfer Protocol](#) (HTTP) used in the [World Wide Web](#)
 - 110: [Post Office Protocol](#) (POP3)
 - 119: [Network News Transfer Protocol](#) (NNTP)
 - 143: [Internet Message Access Protocol](#) (IMAP)
 - 161: [Simple Network Management Protocol](#) (SNMP)
 - 194: [Internet Relay Chat](#) (IRC)
 - 443: [HTTP Secure](#) (HTTPS)
 - 465: SMTP Secure (SMTPS)
 - 8443: [Router](#) remote access



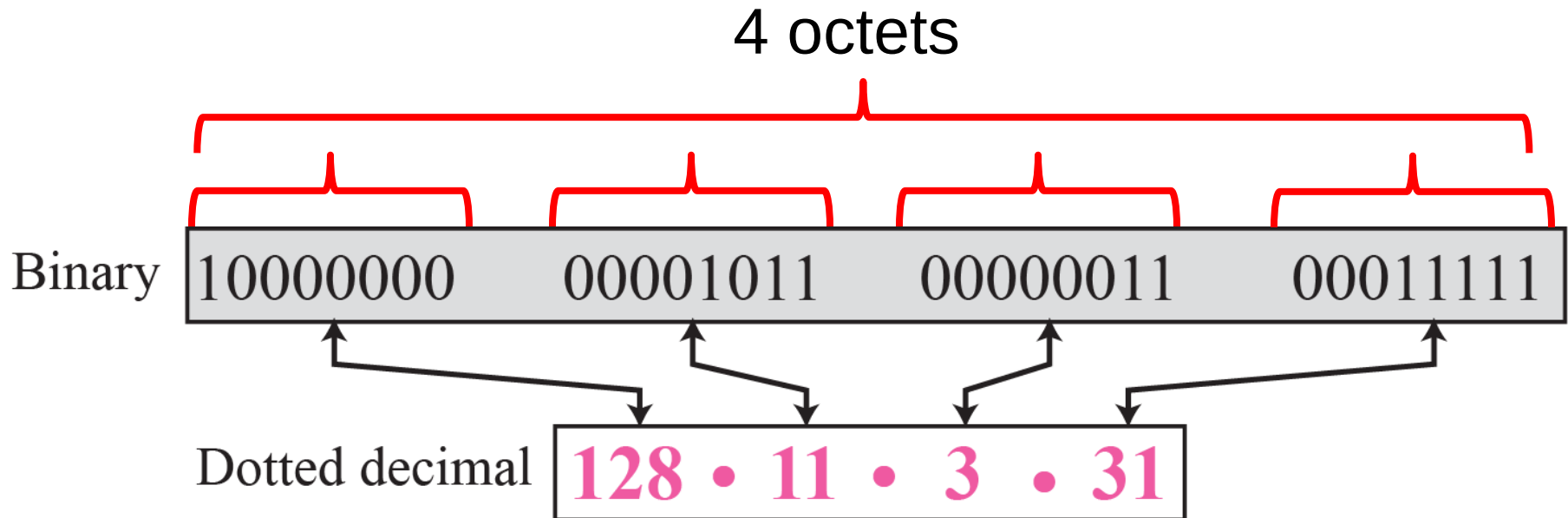
Logical Address

- The identifier used in the network layer of the TCP/IP protocol suite to identify each device connected to the Internet is called the Internet address or **IP address**.
 - IPv4 : 32 bits address
 - IPv6: 128 bits address



Logical Address

- Example of IPv4 address:





Physical Address

- Media Access Control (**MAC**) address, burned-in address (**BIA**), Ethernet hardware address (**EHA**) or hardware address
- A unique identifier assigned to network interfaces for communications on the physical network segment
- Assigned by the manufacturer



Physical Address

- 48-bit (6-byte) written as 12 hexadecimal digits, every byte (2 hexadecimal digits) is separated by a colon.
- Example:

07:01:02:01:2C:4B

Organizational Unique Identifier

Vendor Assigned



Decimal – Hexa - Binary

decimal	hexadecimal	binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Decimal – Hexa - Binary



- Decimal $\leftrightarrow$ Hexa
- Decimal $\leftrightarrow$ Binary
- Binary $\leftrightarrow$ Hexa



IPv4 addressing

	← 32 bits →															
Dotted Decimal	Network Host															
Maximum	255				255				255				255			
Binary	1	8			9	16			17	24			25	32		
	11111111				11111111				11111111				11111111			
	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
Example Decimal	172				16				122				204			
Example Binary	10101100				00010000				01111010				11001100			



IP Classful Addressing

- Class A:

8 bits	8 bits	8 bits	8 bits
Network	Host	Host	Host
- Class B:

Network	Network	Host	Host
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- Class C:

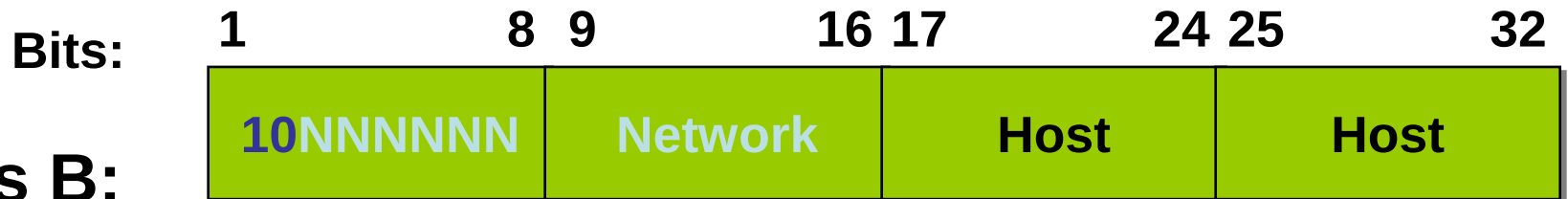
Network	Network	Network	Host
---------	---------	---------	------
- Class D: Multicast
- Class E: Research



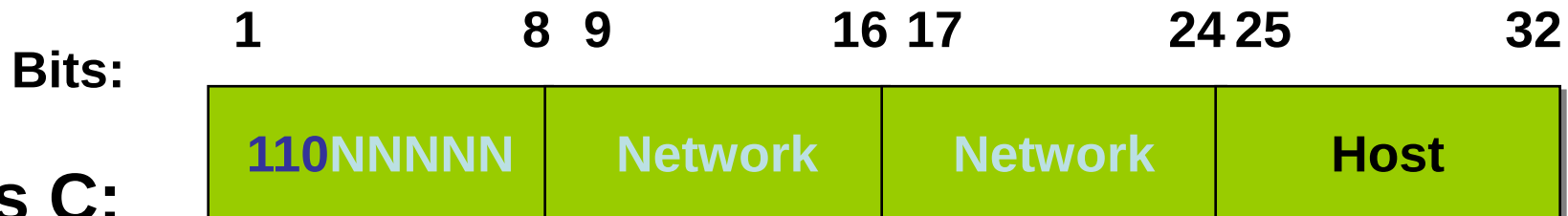
IP address class



Range (0-127)



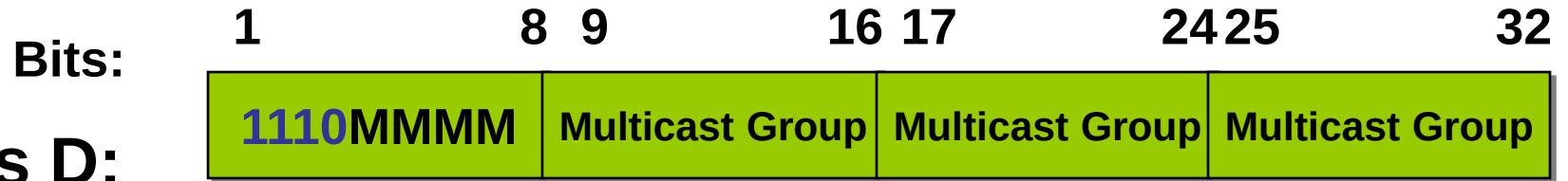
Range (128-191)



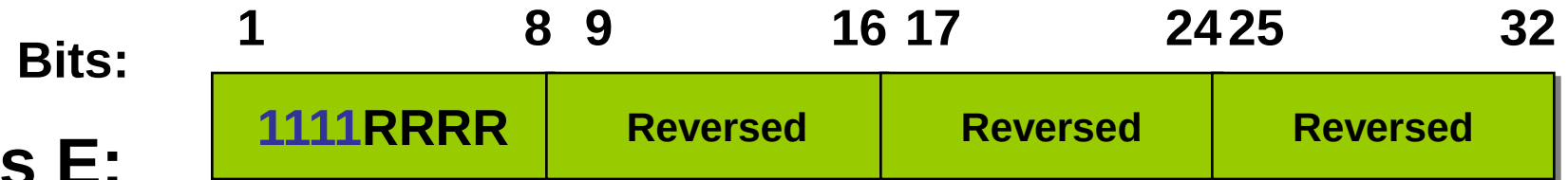
Range (192-223)



IP address class



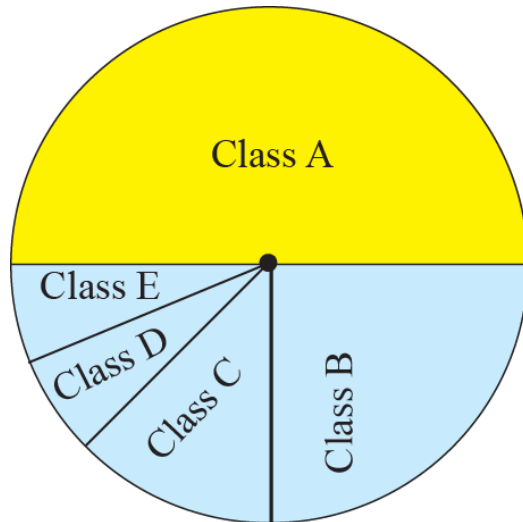
Range (224-239)



Range (240-255)



IP Classful Addressing



Class A: $2^{31} = 2,147,483,648$ addresses, 50%

Class B: $2^{30} = 1,073,741,824$ addresses, 25%

Class C: $2^{29} = 536,870,912$ addresses, 12.5%

Class D: $2^{28} = 268,435,456$ addresses, 6.25%

Class E: $2^{28} = 268,435,456$ addresses, 6.25%



Example

-
- Find the class of each class



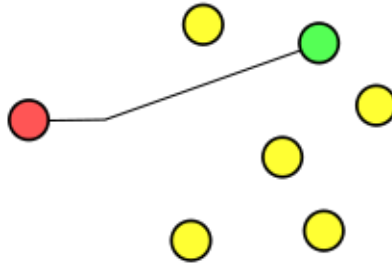
Network & broadcast address

- Network address
 - All host bits are '0'
 - Used by router
- Broadcast address
 - All host bits are '1'
 - Send to all host in the network
- Host address
 - Others
 - Assigned to a host

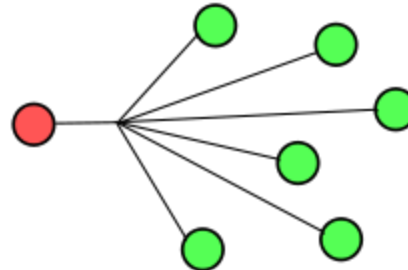


Network & broadcast address

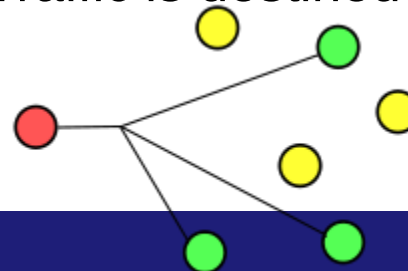
- Unicast (point-to-point) - Unicast Traffic is destined for a single host



- Broadcast: (to all) - Broadcast Traffic is destined for all hosts on a given network

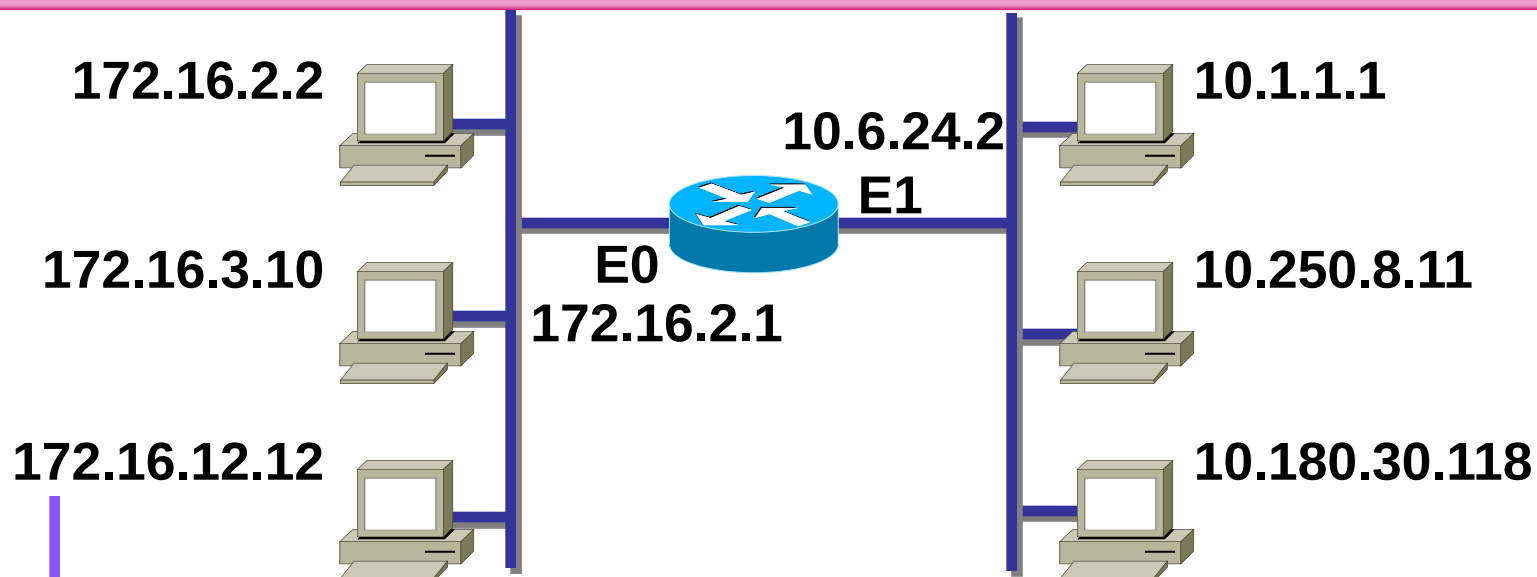


- Multicast: (to a group) Multicast Traffic is destined for a group of hosts requesting multicast





Host address



172.16.12.12

↓

172.16 . 12 . 12

Network Host

Routing Table	
Network	Interface
172.16.0.0	E0
10.0.0.0	E1



Network		Host																
172	16	0								0								
		16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	n
10101100	00010000	00000000								00000000								1
		00000000								00000001								2
		00000000								00000011								3
		⋮								⋮								⋮
		11111111								11111101								65534
		11111111								11111110								65535
		11111111								11111111								65536
																		- 2
		$2^n - 2 = 2^{16} - 2 = 65534$																
		65534																



Example

- ❖ 125.0.0.0
- ❖ 128.15.0.0
- ❖ 190.0.0.0
- ❖ 192.168.0.0
- ❖ 195.169.1.0



Private & public address

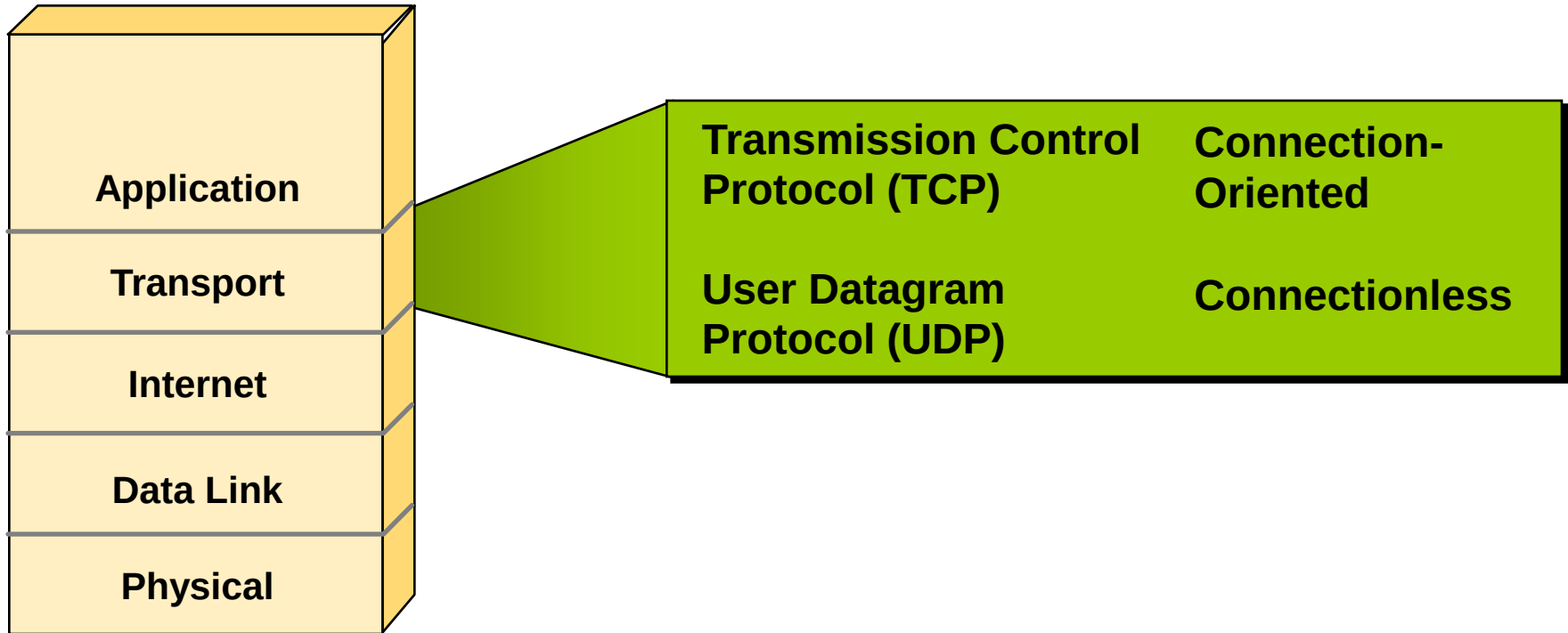


Table 1-2 *Private IP Addresses*

Class	RFC 1918 Internal Address Range
A	10.0.0.0 to 10.255.255.255
B	172.16.0.0 to 172.31.255.255
C	192.168.0.0 to 192.168.255.255



Transport Layer Overview





TCP Segment Format

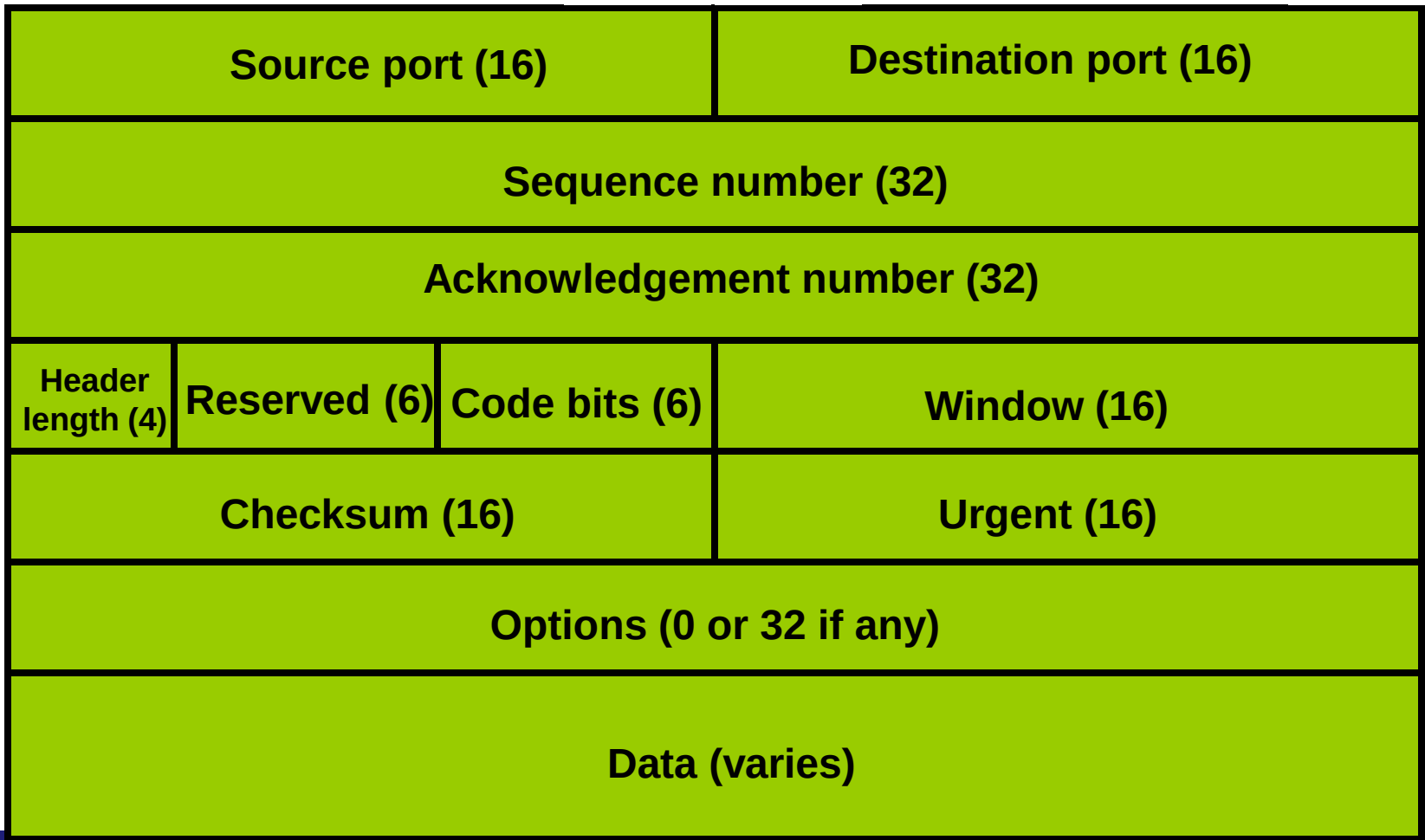


Bit 0

Bit 15

Bit 16

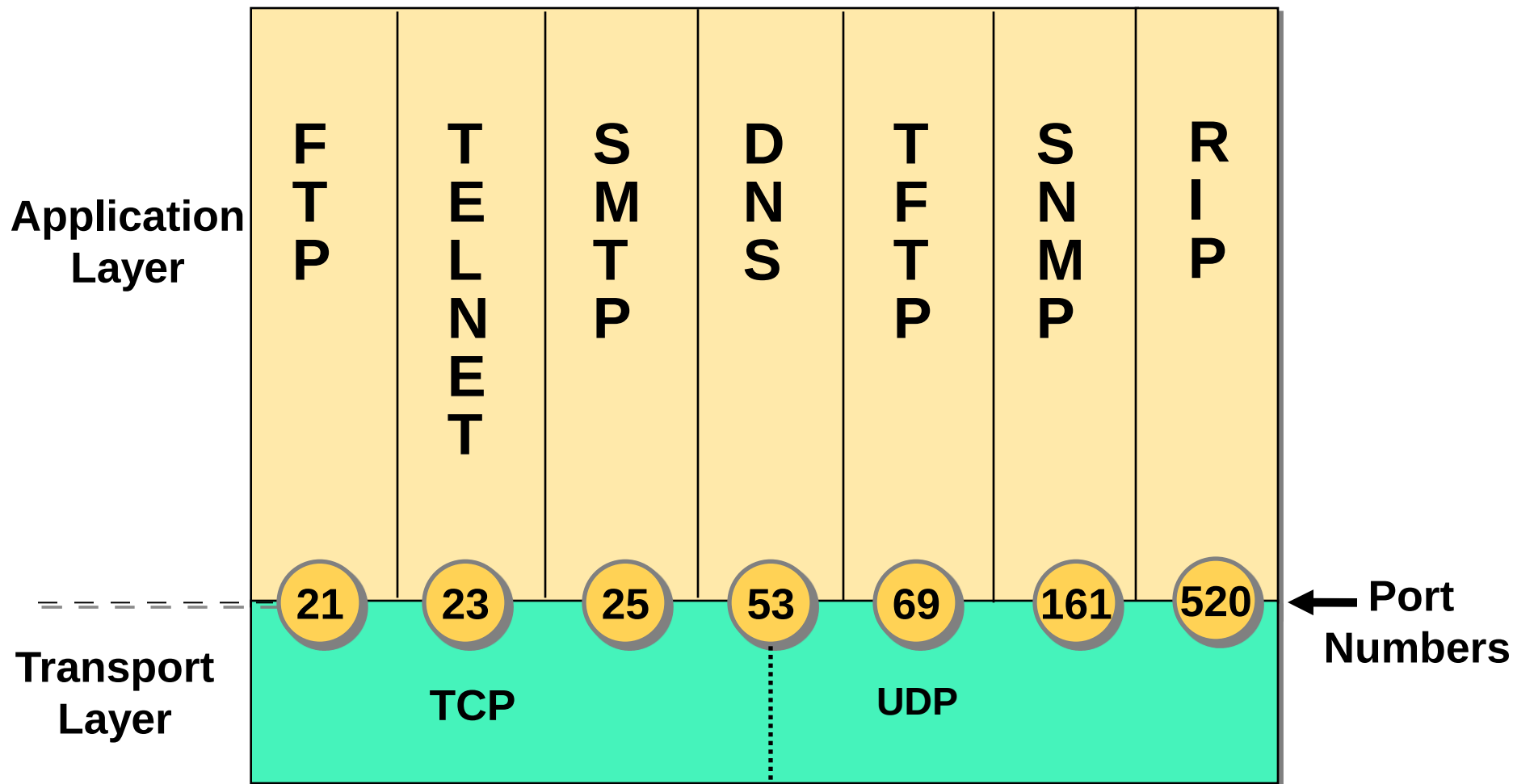
Bit 31



20
Bytes

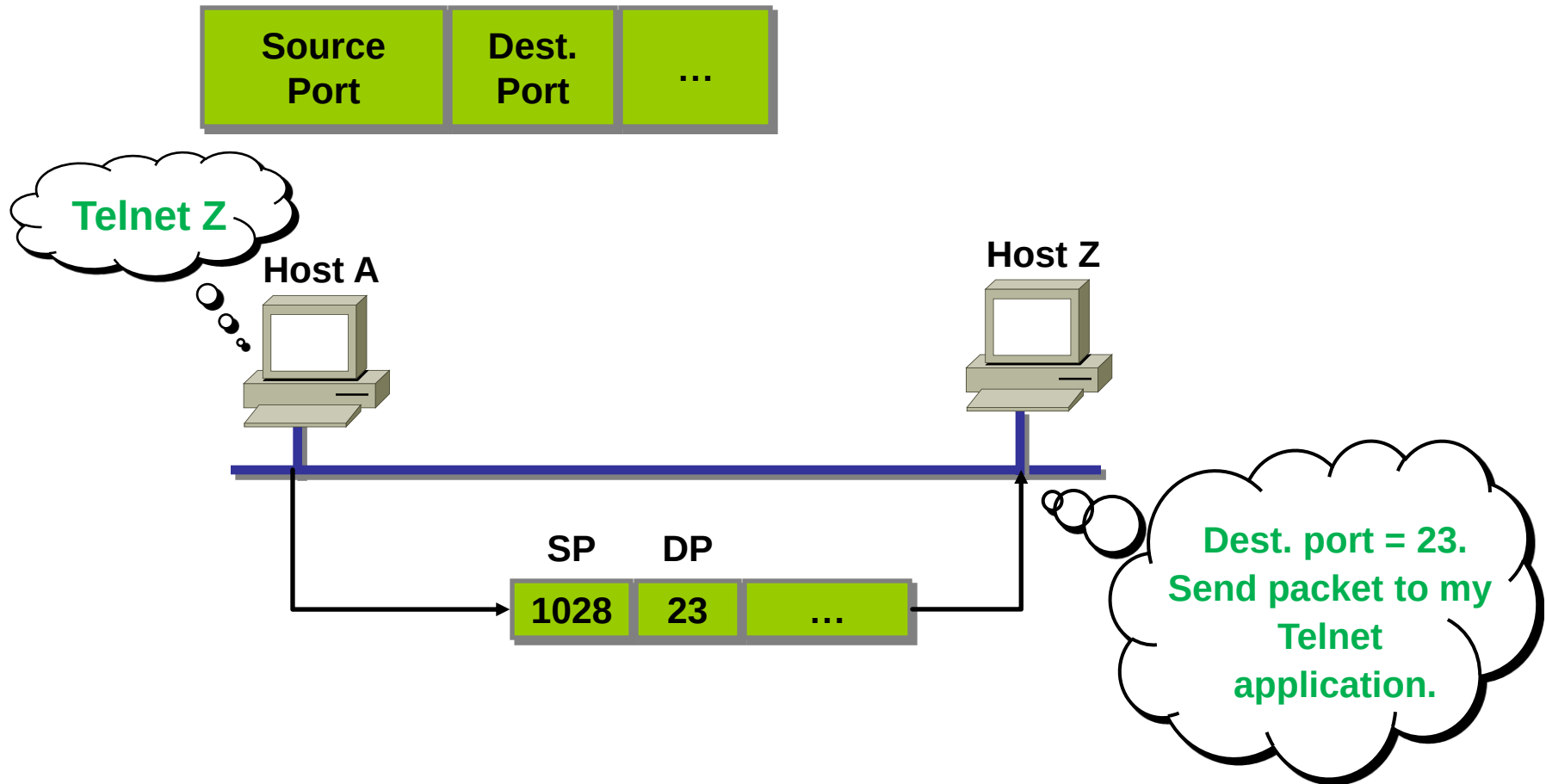


Port Numbers



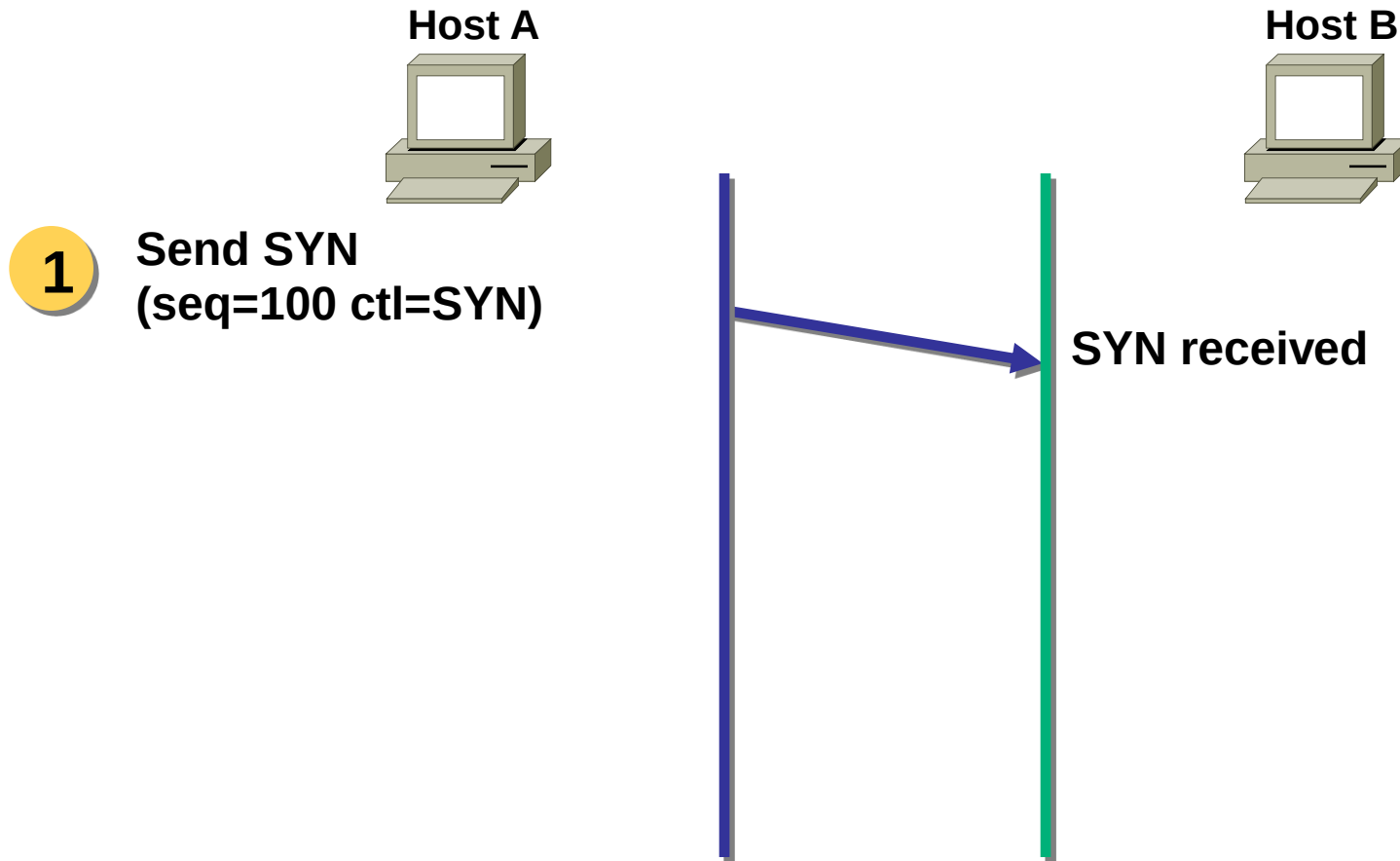


TCP Port Numbers



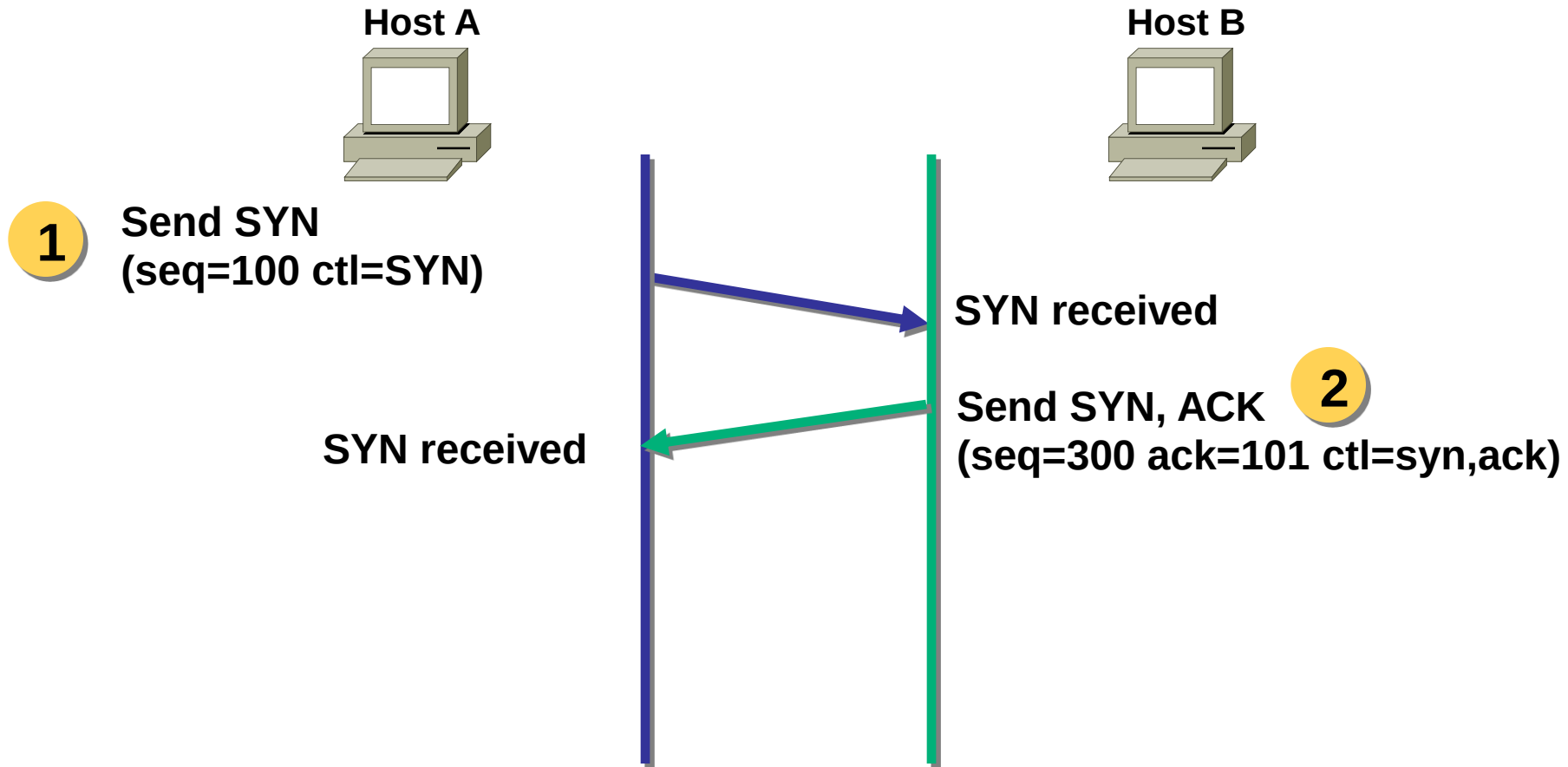


TCP Three Way Handshake/Open Connection



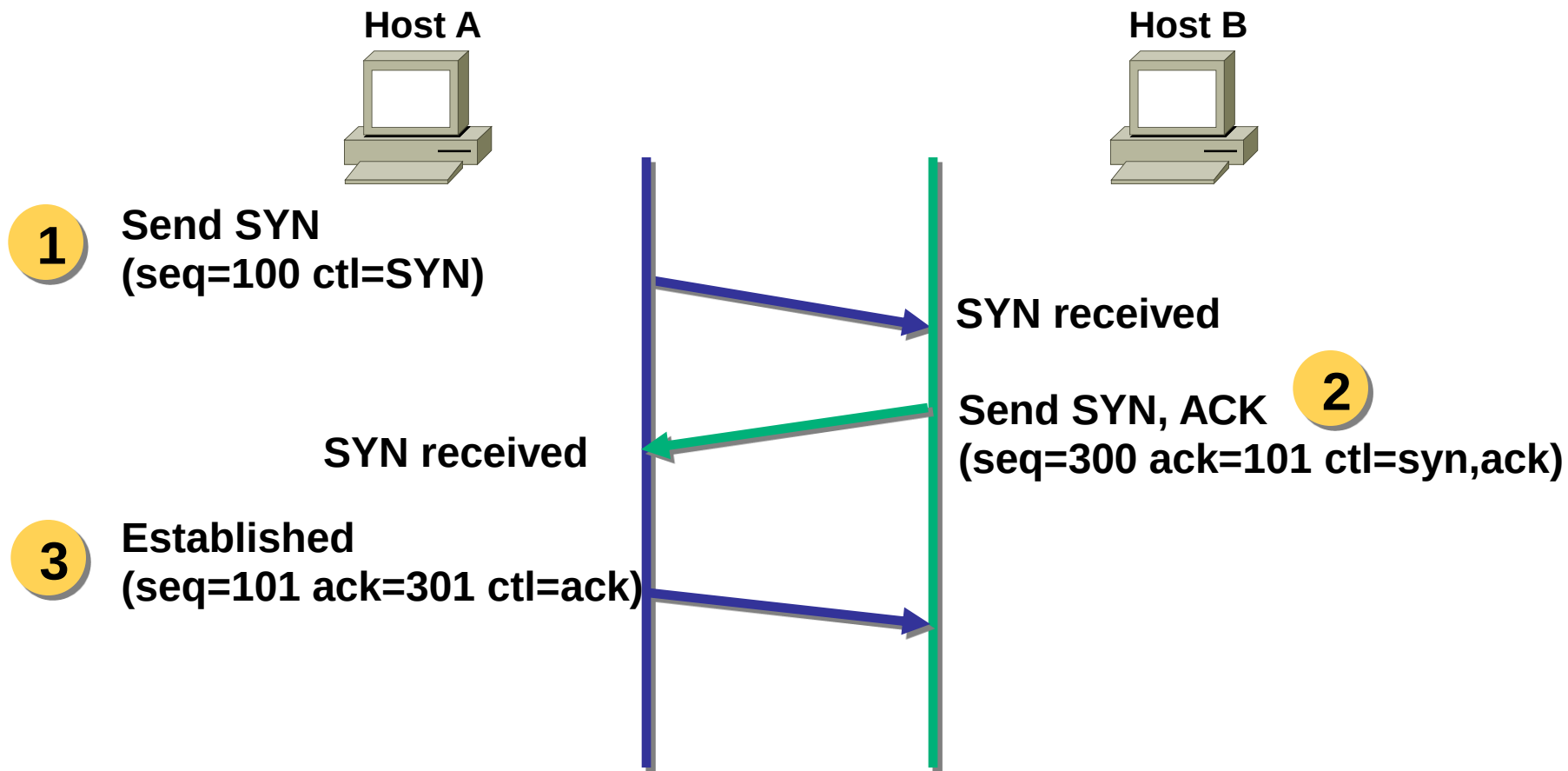


TCP Three Way Handshake/Open Connection





TCP Three Way Handshake/Open Connection





TCP Simple Acknowledgment

Sender

Receiver

- Window size = 1



TCP Simple Acknowledgment

Sender

Receiver

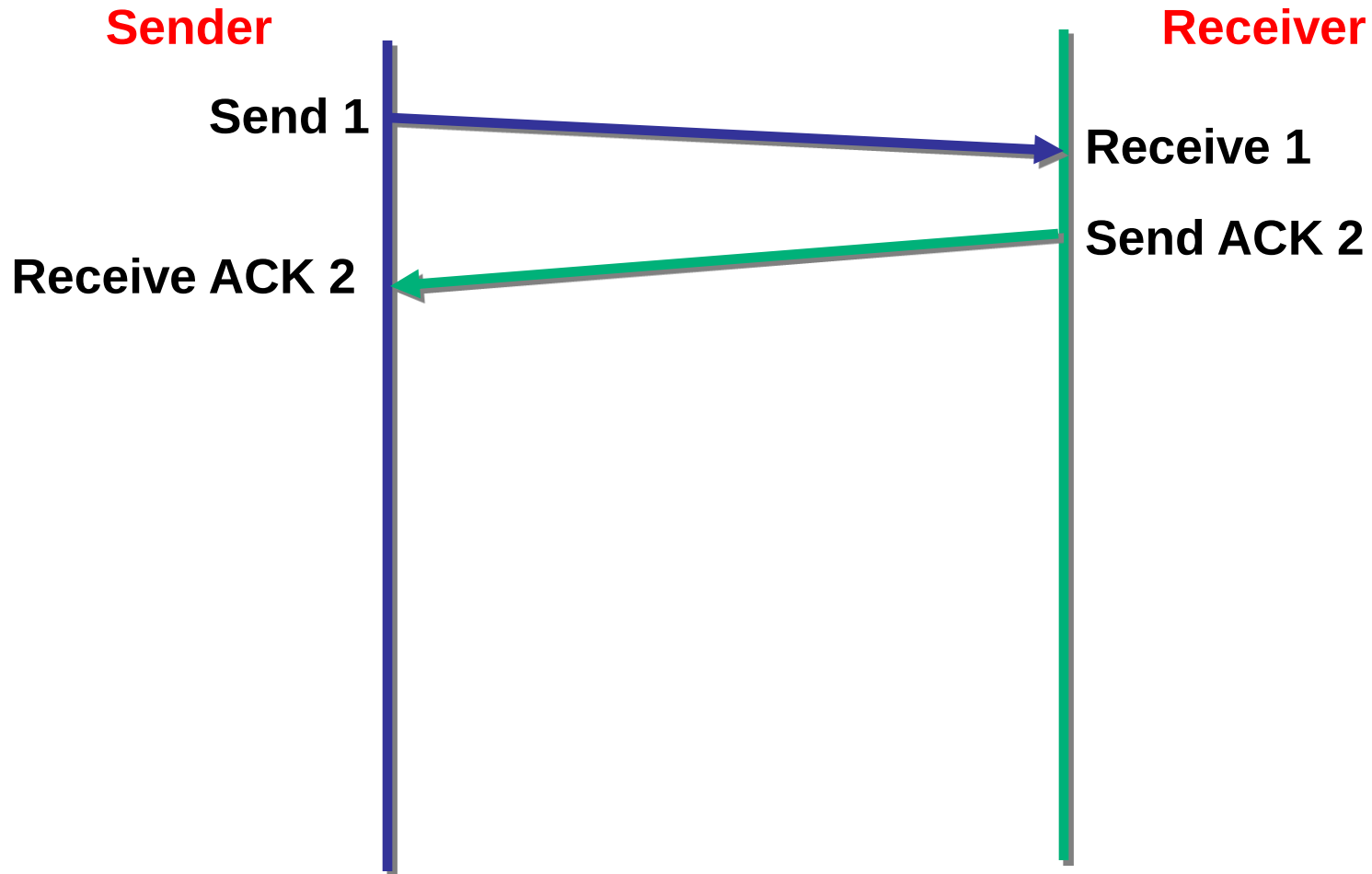
Send 1

Receive 1

- Window size = 1



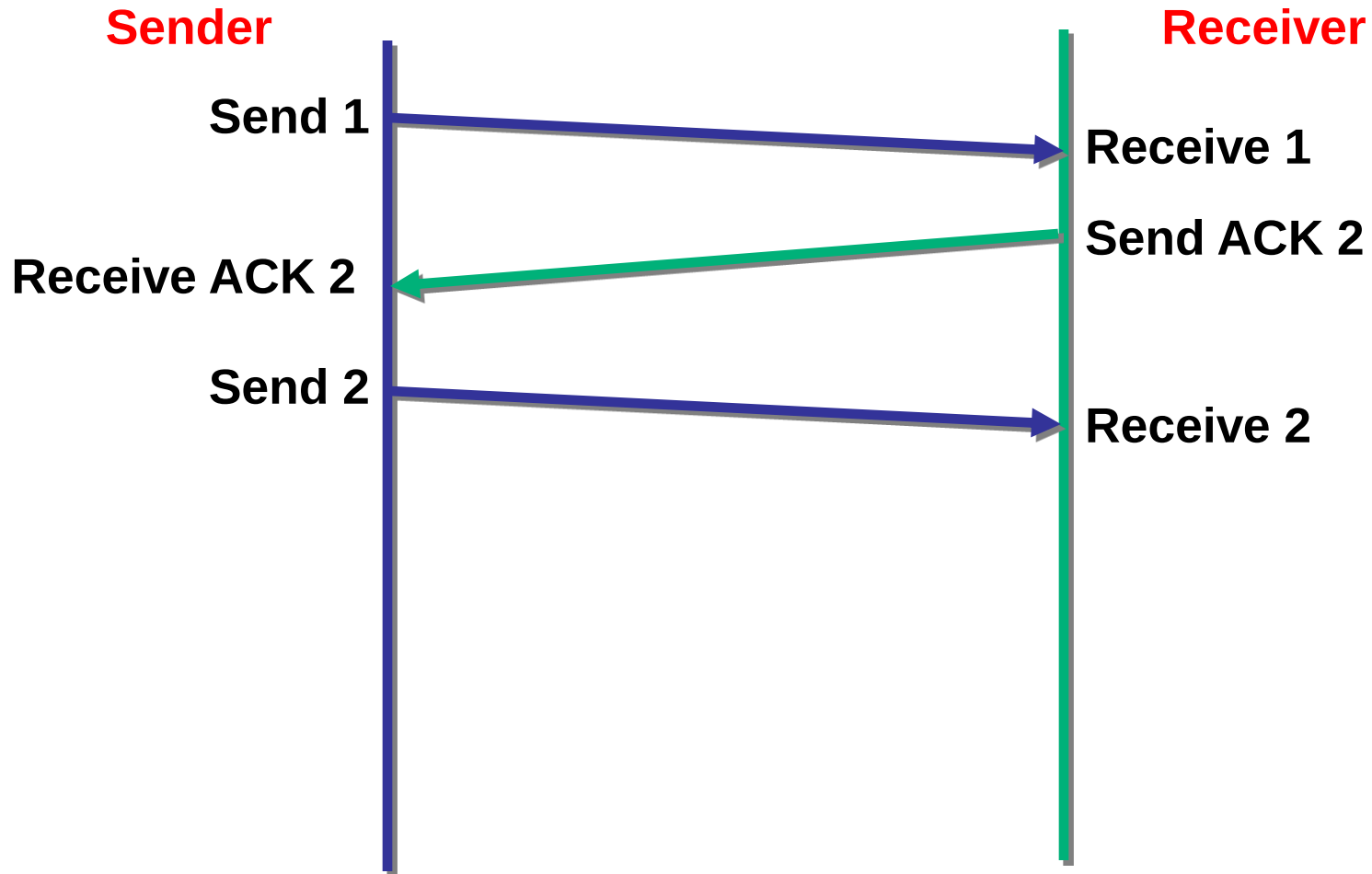
TCP Simple Acknowledgment



- Window size = 1



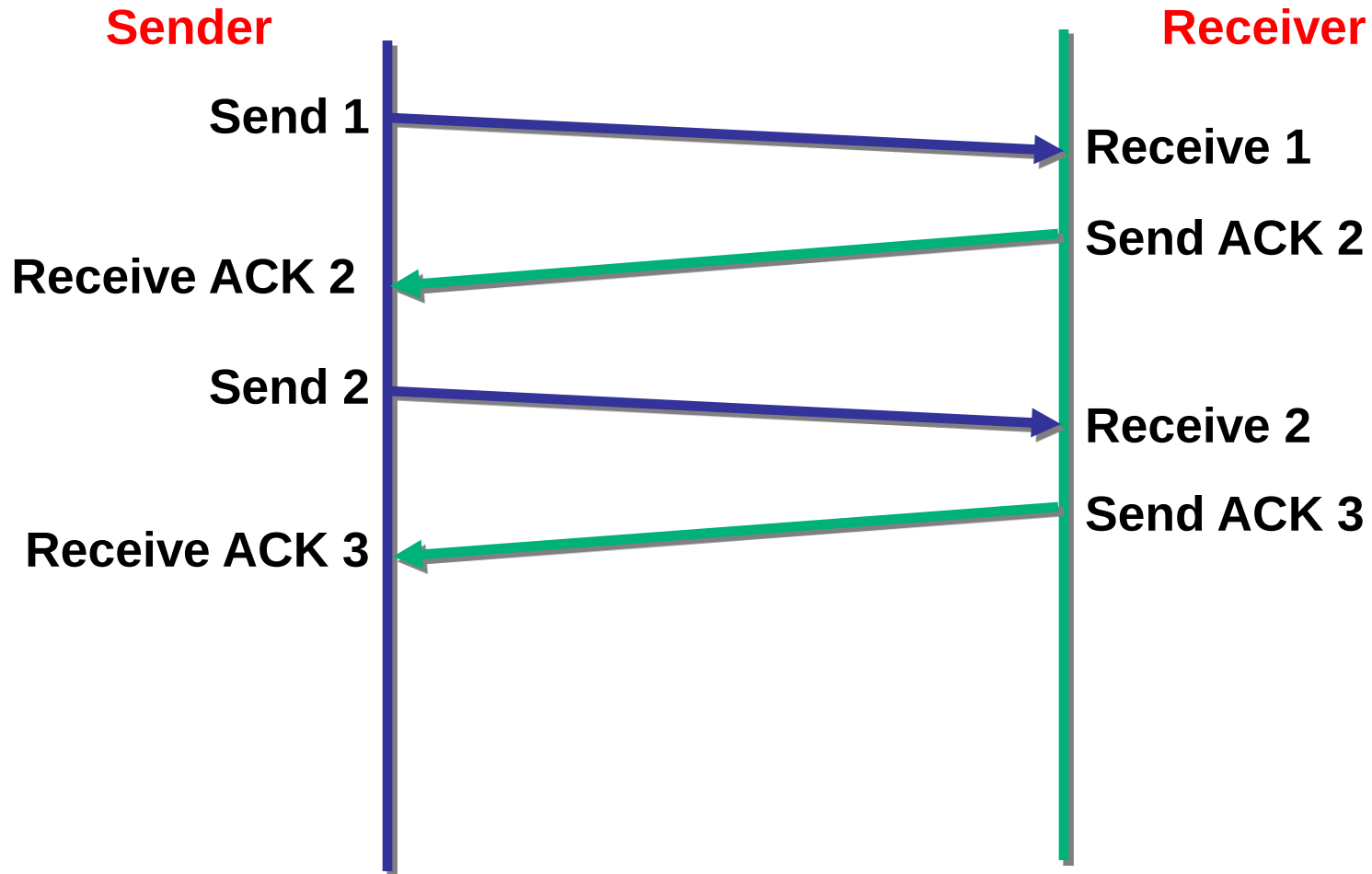
TCP Simple Acknowledgment



■ Window size = 1



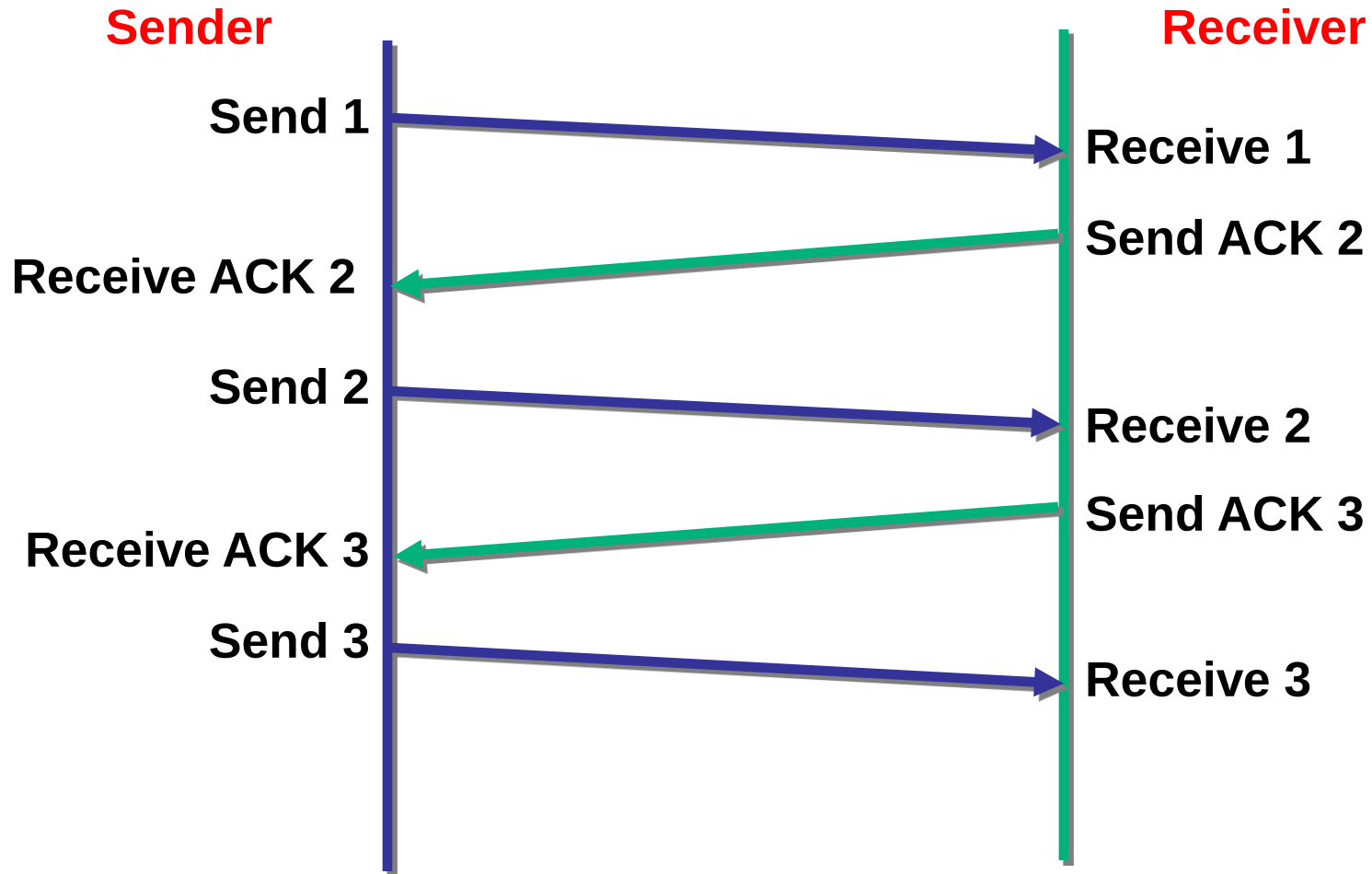
TCP Simple Acknowledgment



■ Window size = 1



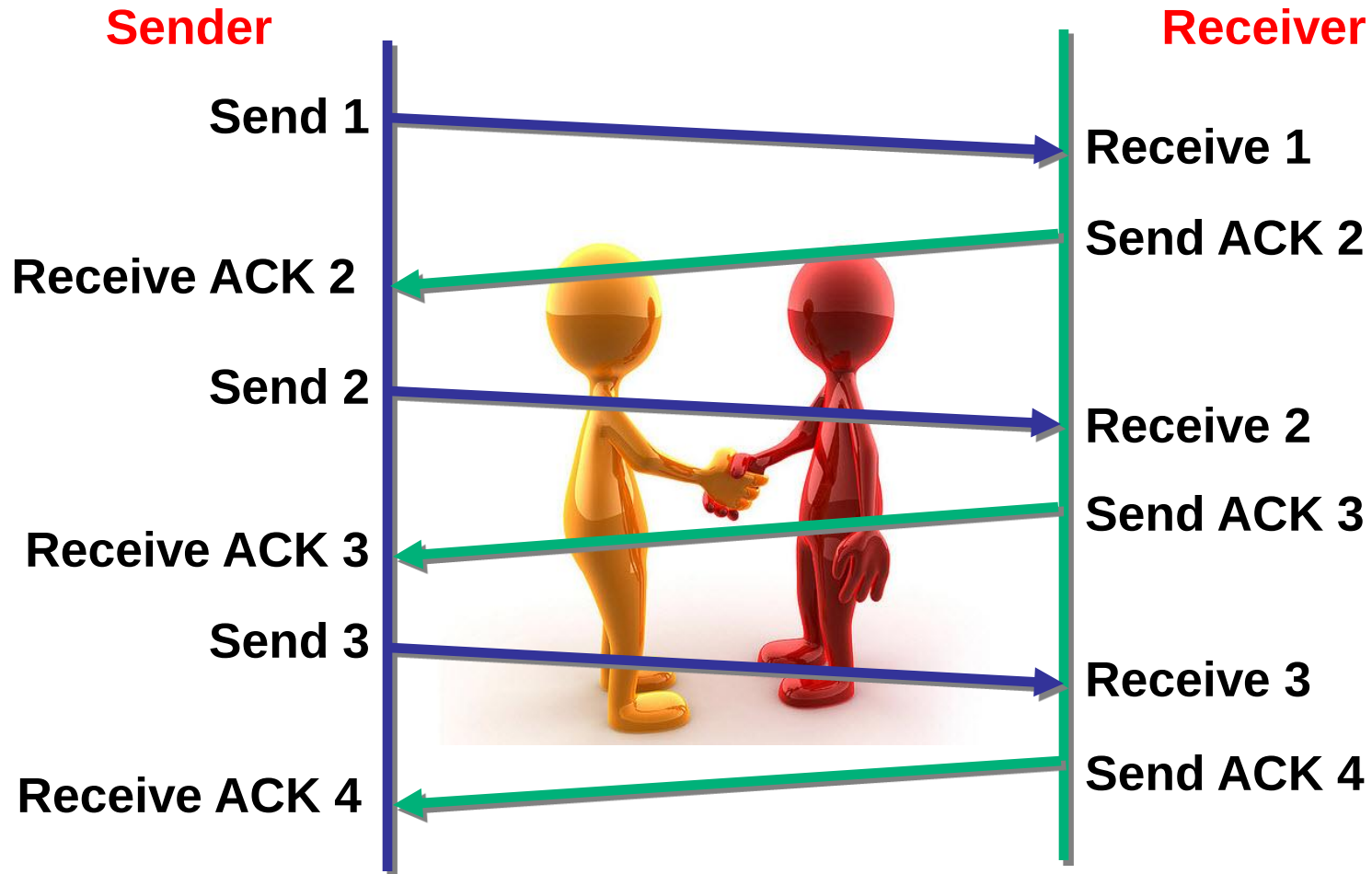
TCP Simple Acknowledgment



■ Window size = 1



TCP Simple Acknowledgment



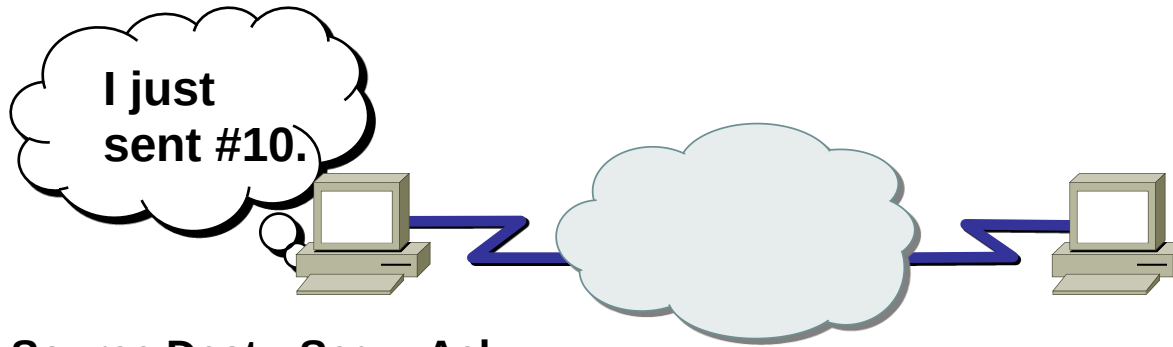
■ Window size = 1



TCP Sequence and Acknowledgment Numbers



Source Port	Dest. Port	Sequence #	Acknowledgement #	...
-------------	------------	------------	-------------------	-----



Source	Dest.	Seq.	Ack.
1028	23	10	1

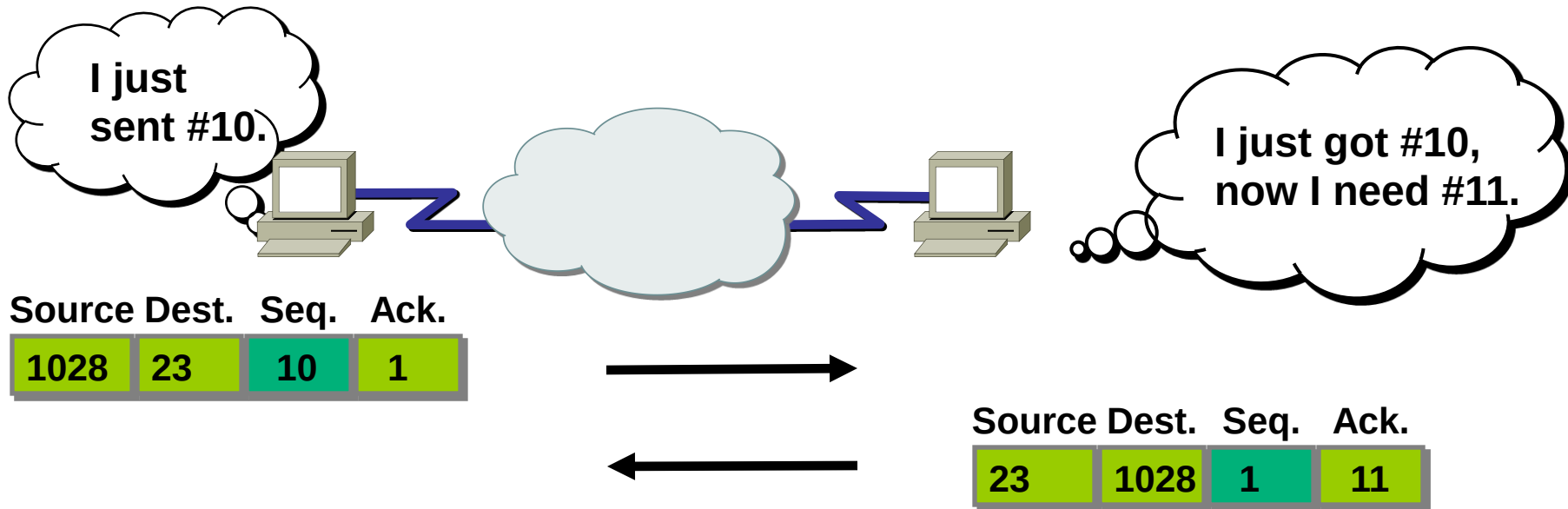




TCP Sequence and Acknowledgment Numbers



Source Port	Dest. Port	Sequence #	Acknowledgement #	...
-------------	------------	------------	-------------------	-----

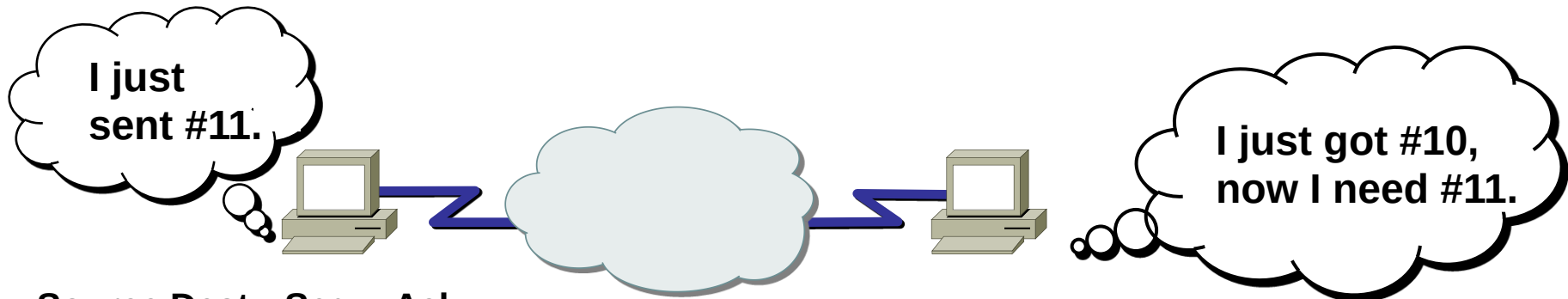




TCP Sequence and Acknowledgment Numbers



Source Port	Dest. Port	Sequence #	Acknowledgement #	...
-------------	------------	------------	-------------------	-----



Source	Dest.	Seq.	Ack.
1028	23	10	1



Source	Dest.	Seq.	Ack.
23	1028	1	11

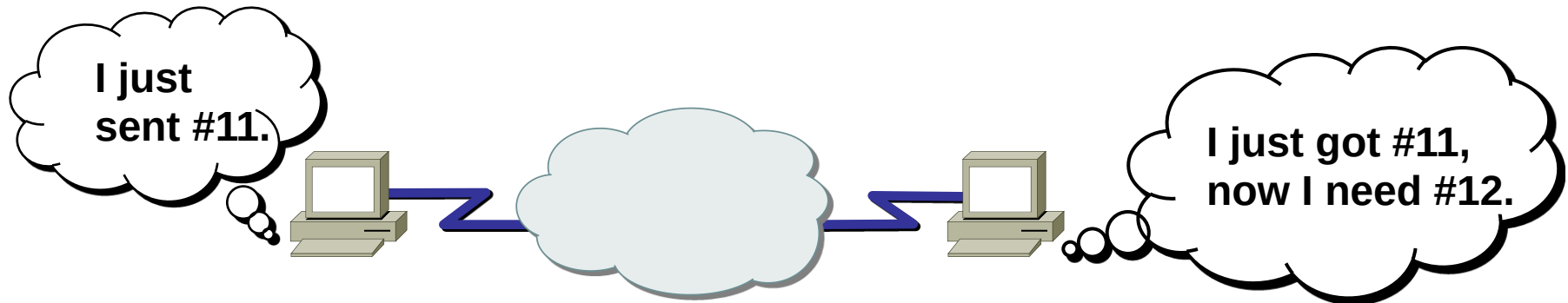
Source	Dest.	Seq.	Ack.
1028	23	11	2



TCP Sequence and Acknowledgment Numbers



Source Port	Dest. Port	Sequence #	Acknowledgement #	...
-------------	------------	------------	-------------------	-----



1028	23	10	1
------	----	----	---



Source Dest. Seq. Ack.

1028	23	11	2
------	----	----	---

Source Dest. Seq. Ack.

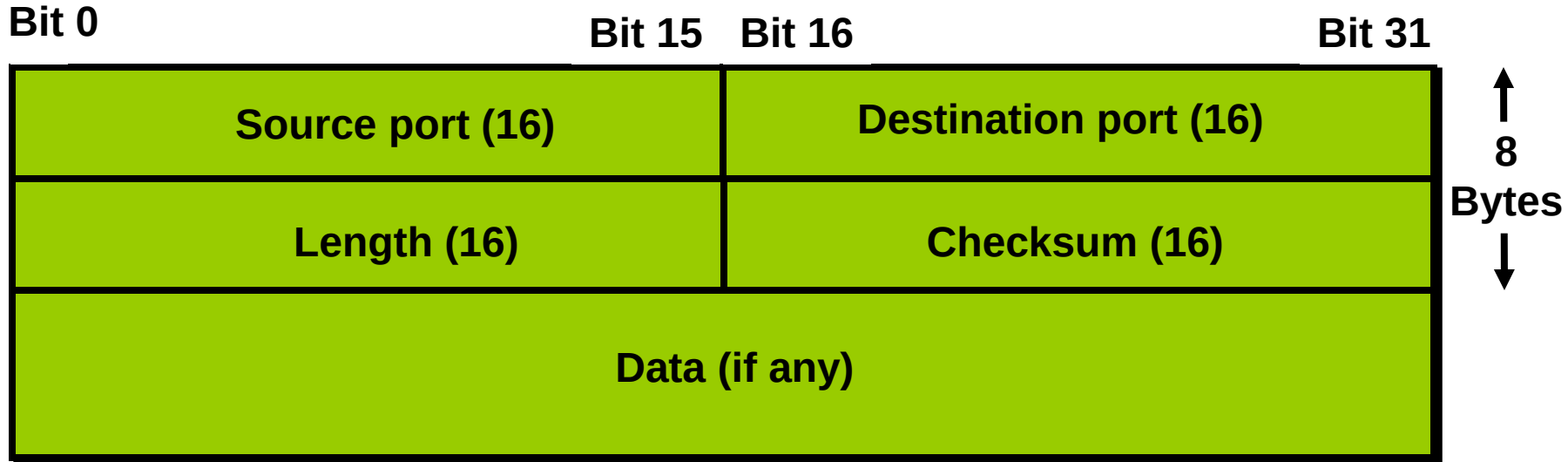
23	1028	1	11
----	------	---	----

Source Dest. Seq. Ack.

23	1028	2	12
----	------	---	----



UDP Segment Format



- No sequence or acknowledgment fields



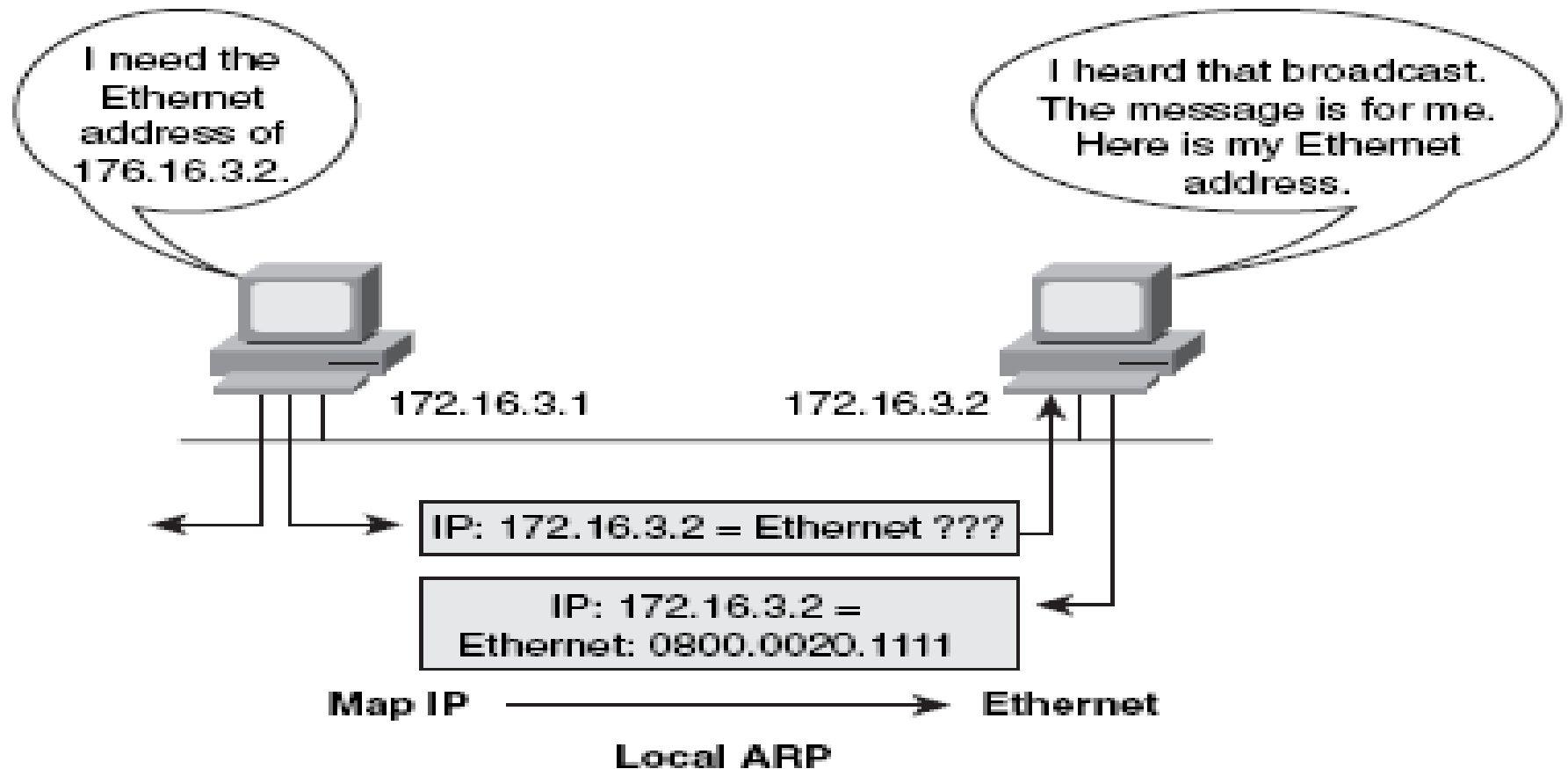
Mapping L2 Address to L3 Address



- To send data to a destination, a host must know the physical (MAC) address of the destination
- Address Resolution Protocol (ARP)
 - Binding IP address – MAC address
 - ARP request, ARP reply
 - ARP table aging time

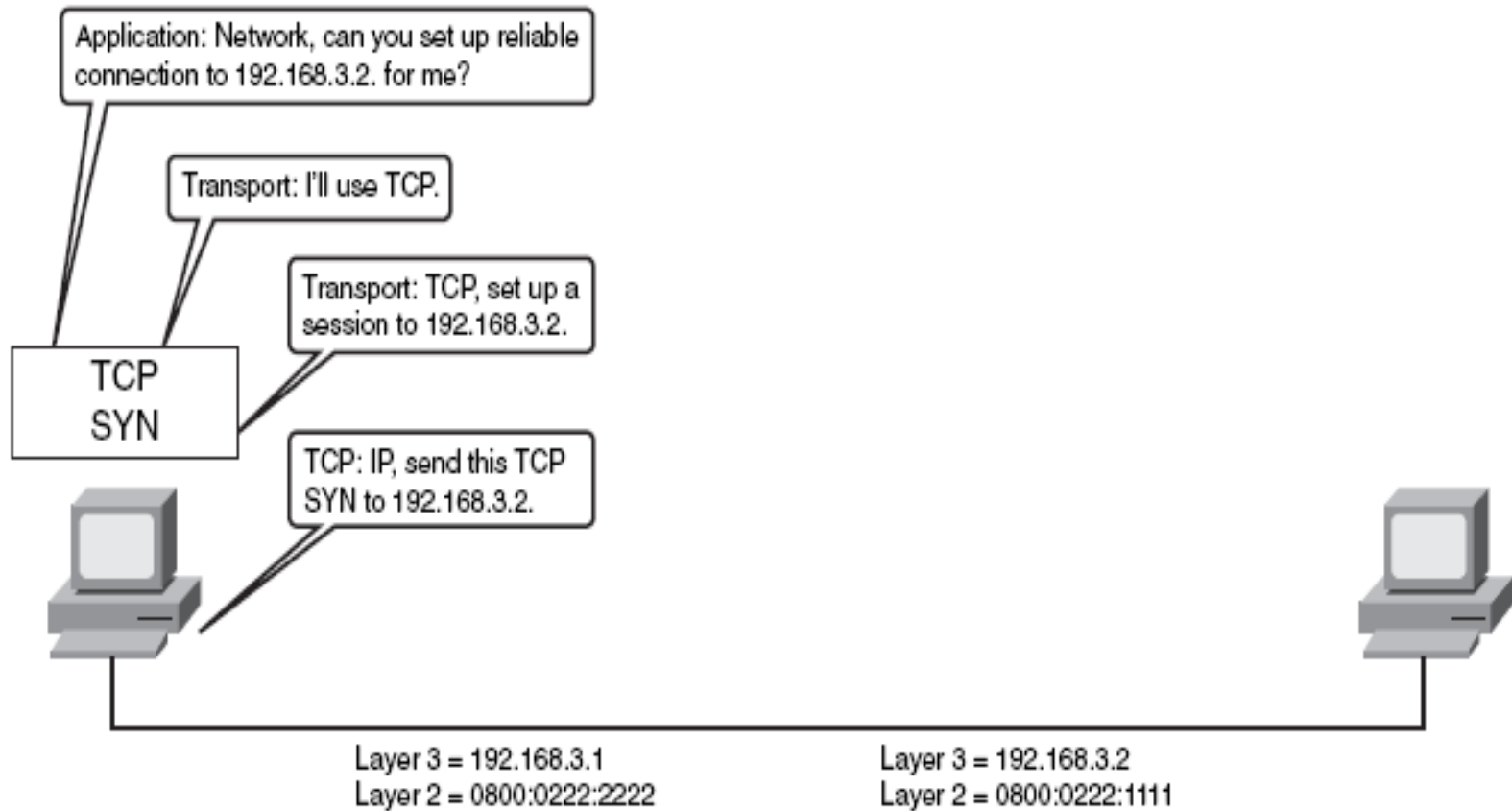


Mapping L2 Address to L3 Address



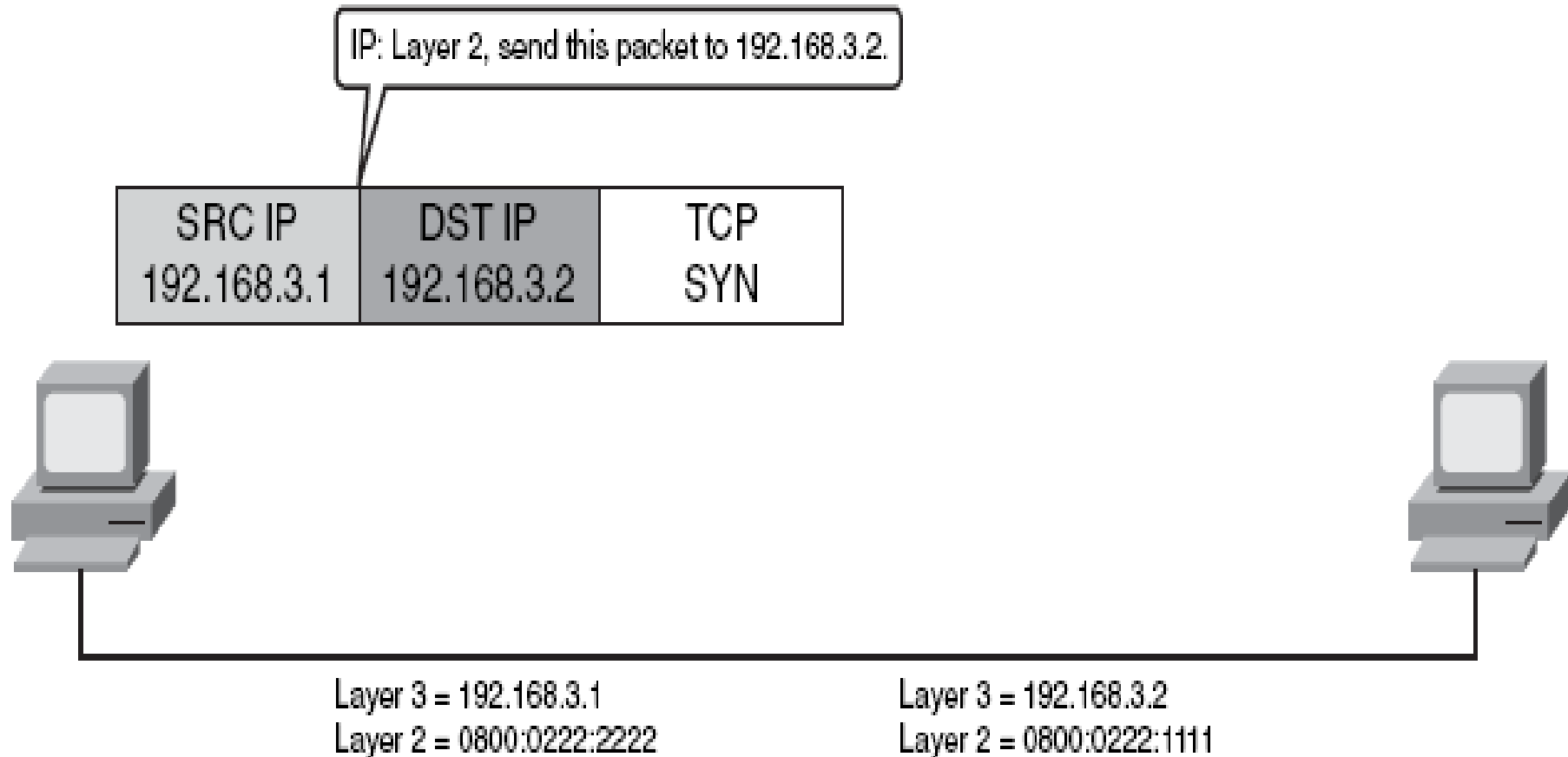


Packet delivery



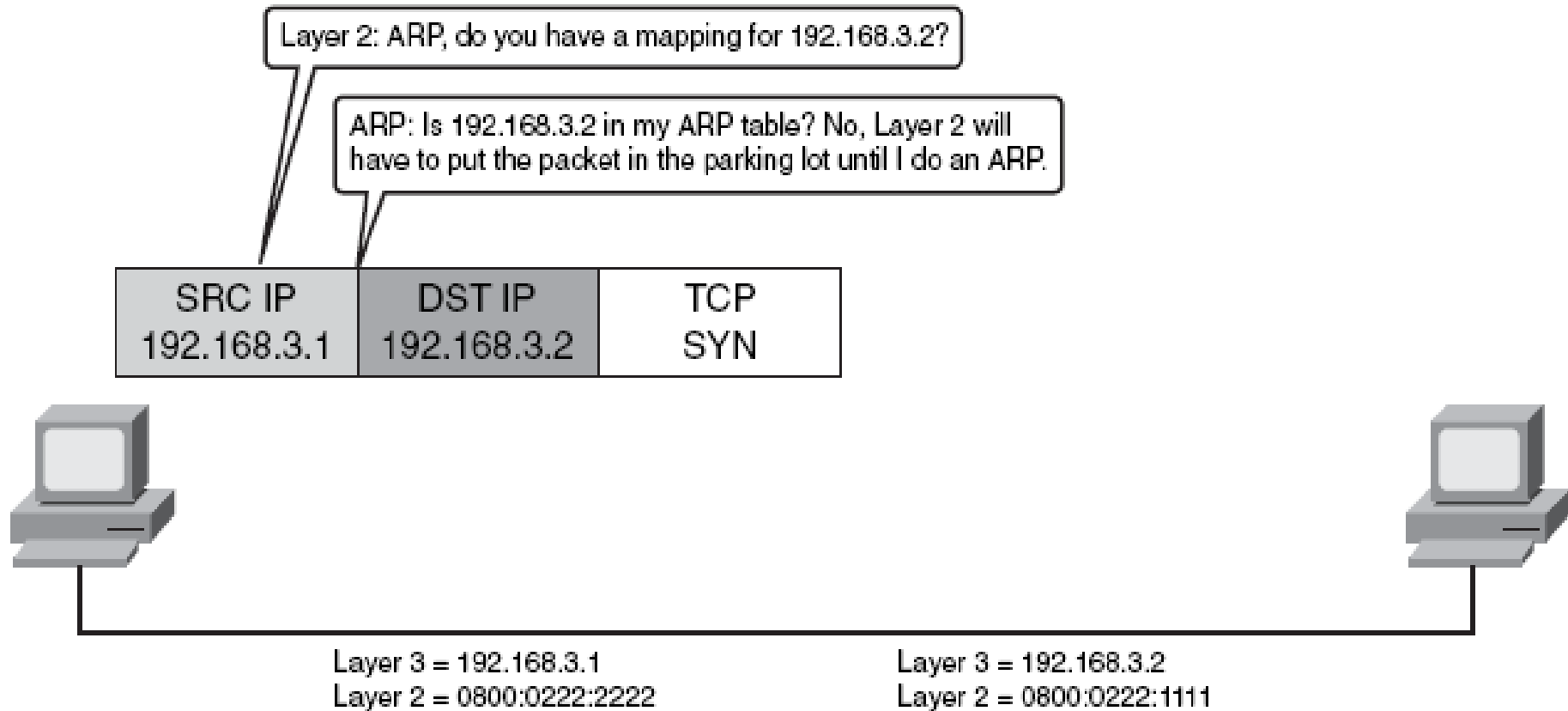


IP Layer Operation



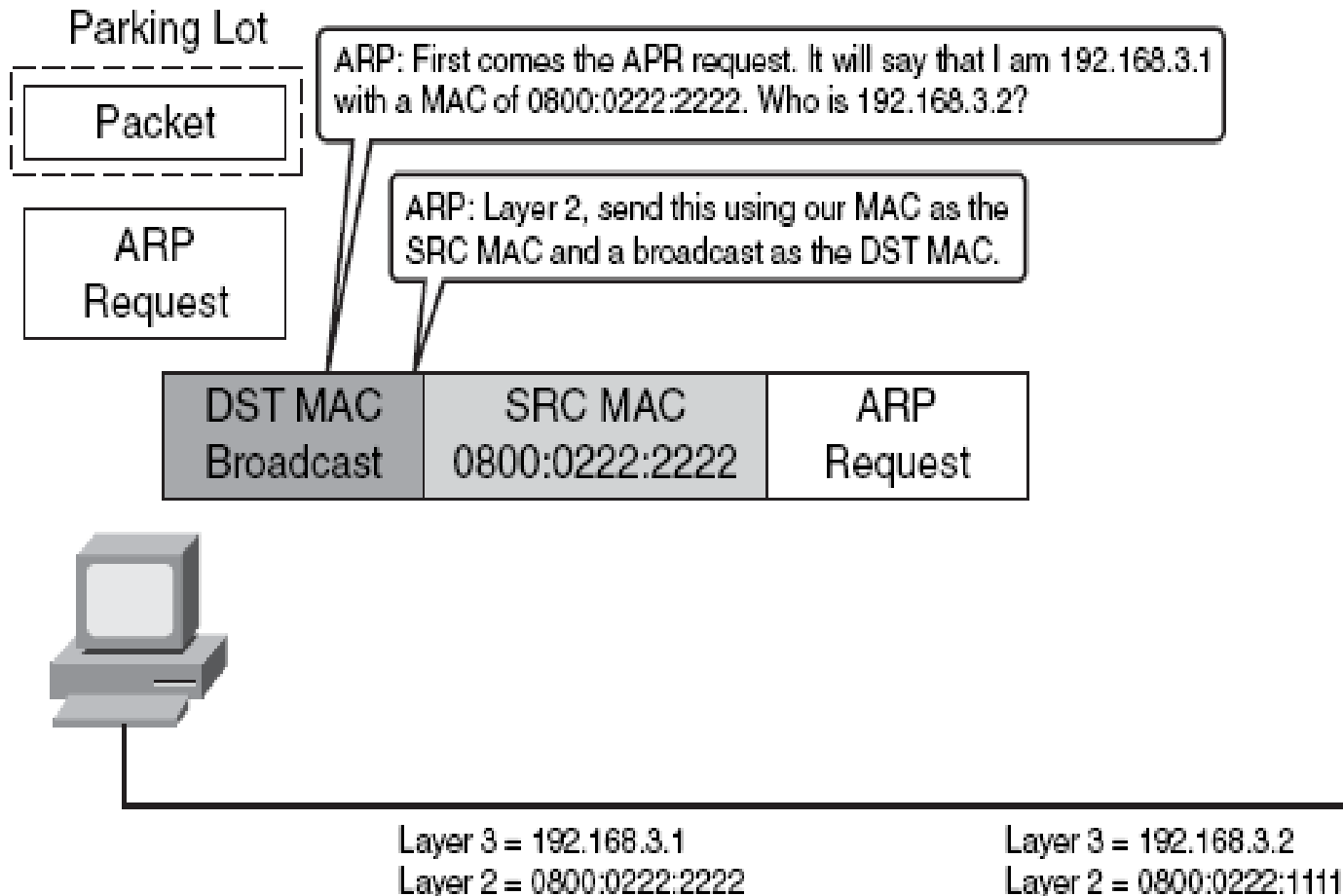


ARP table lookup



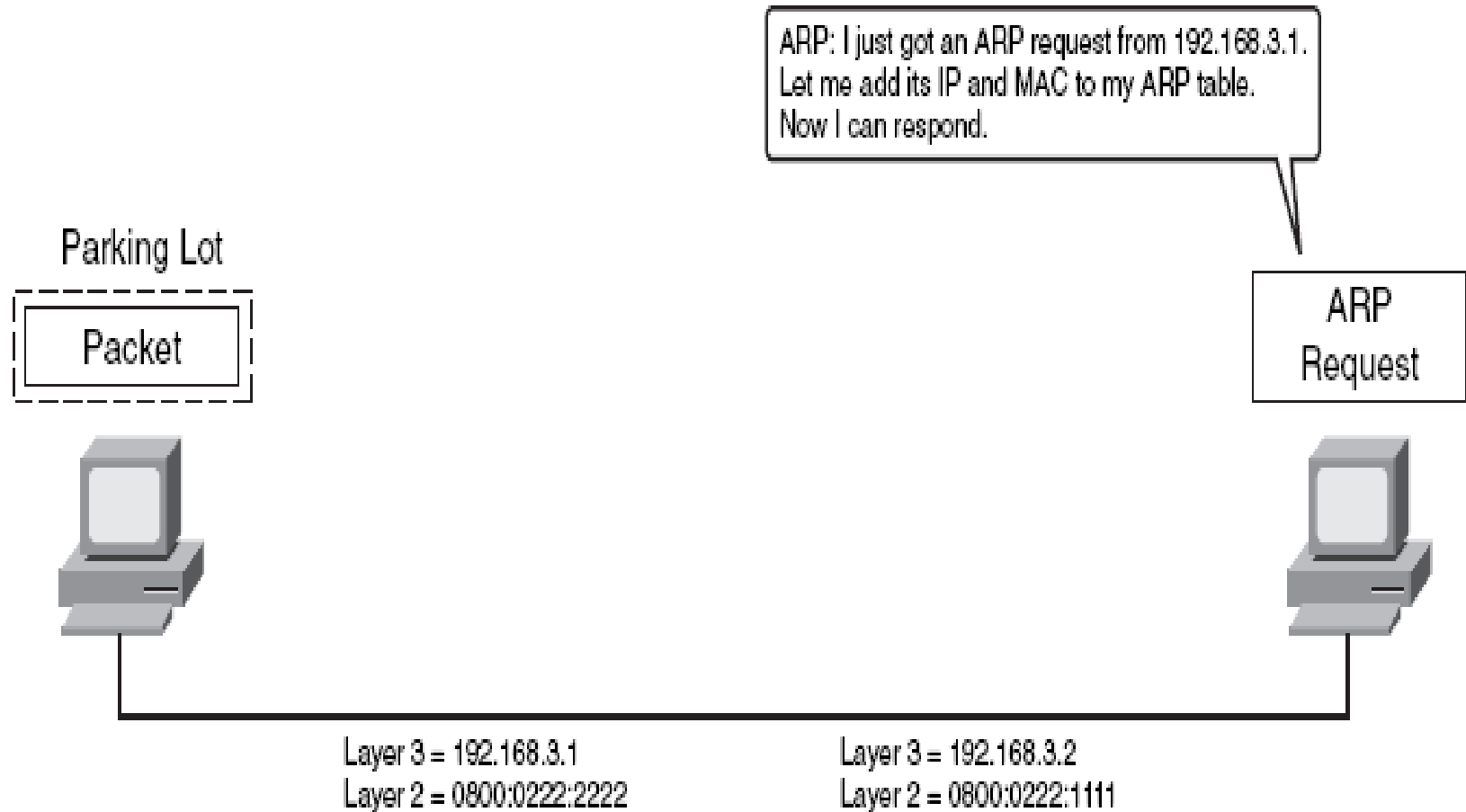


ARP request



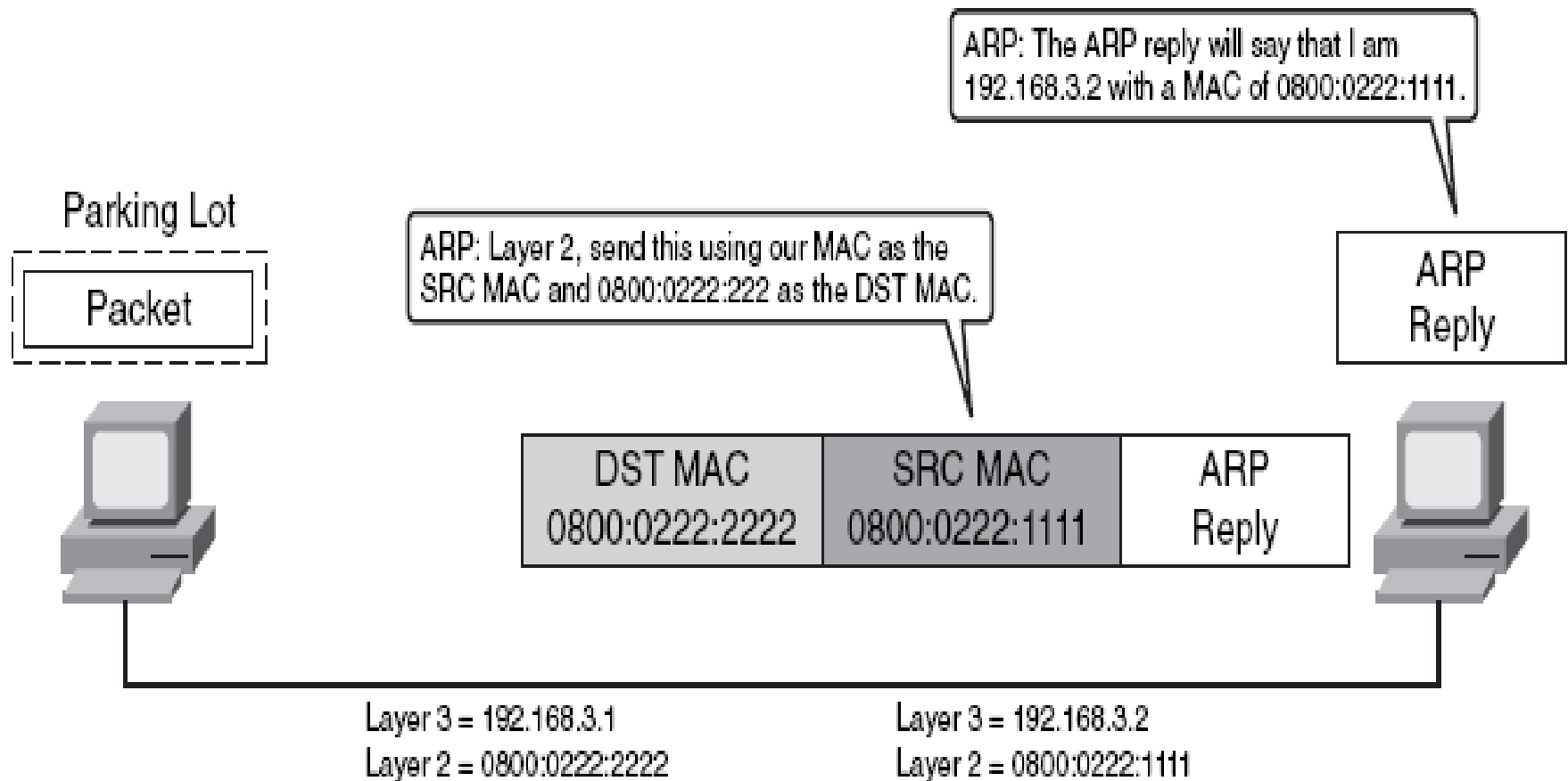


ARP learning



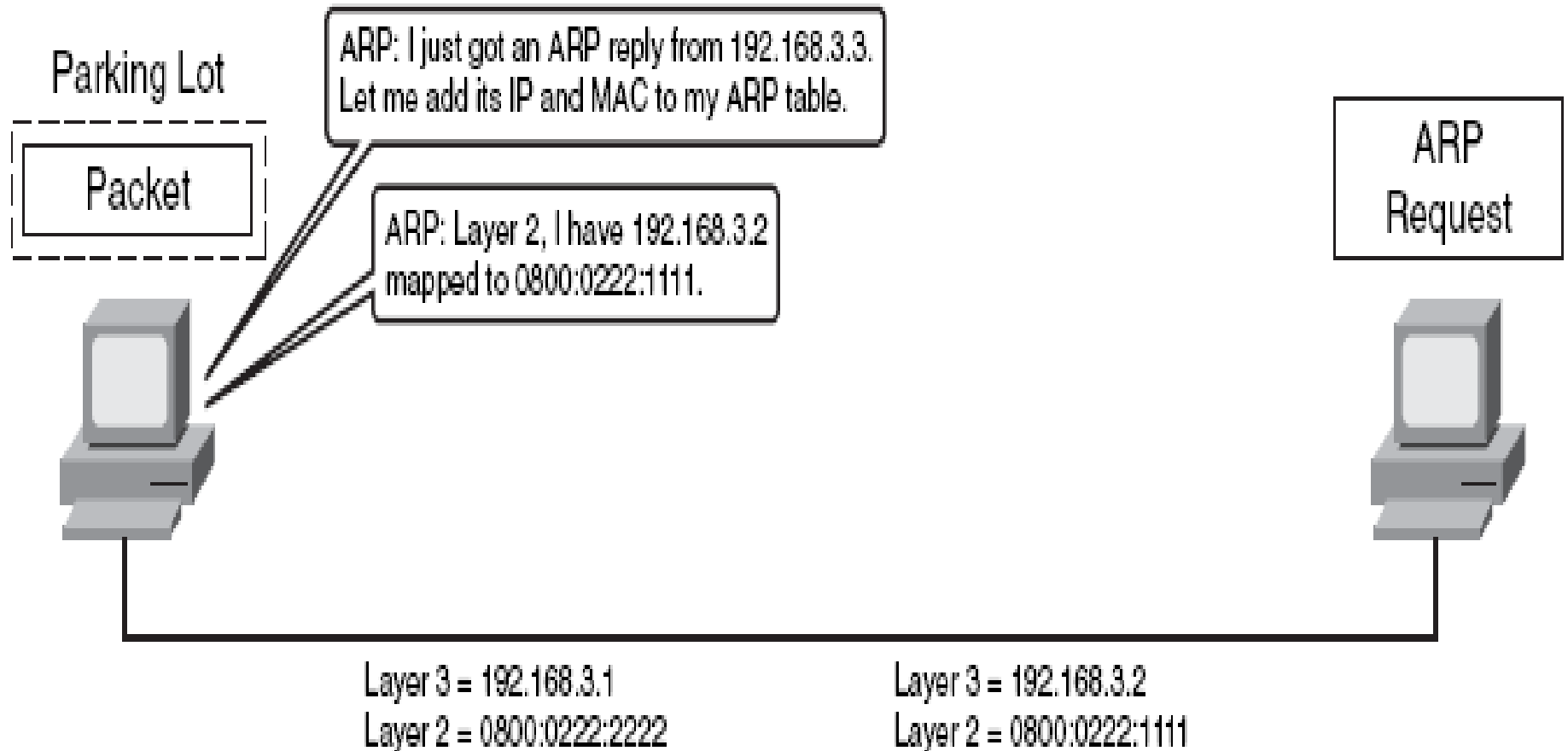


ARP reply



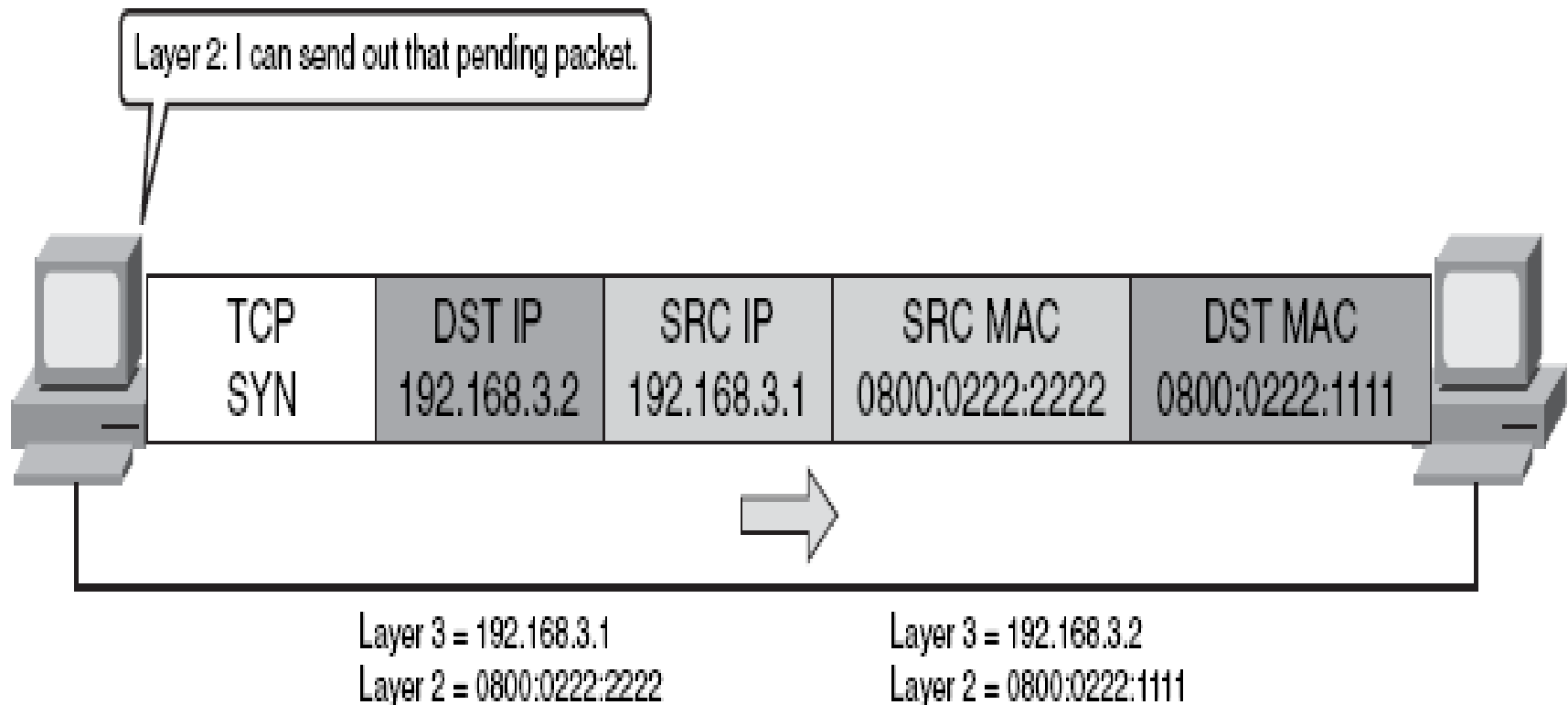


ARP update table



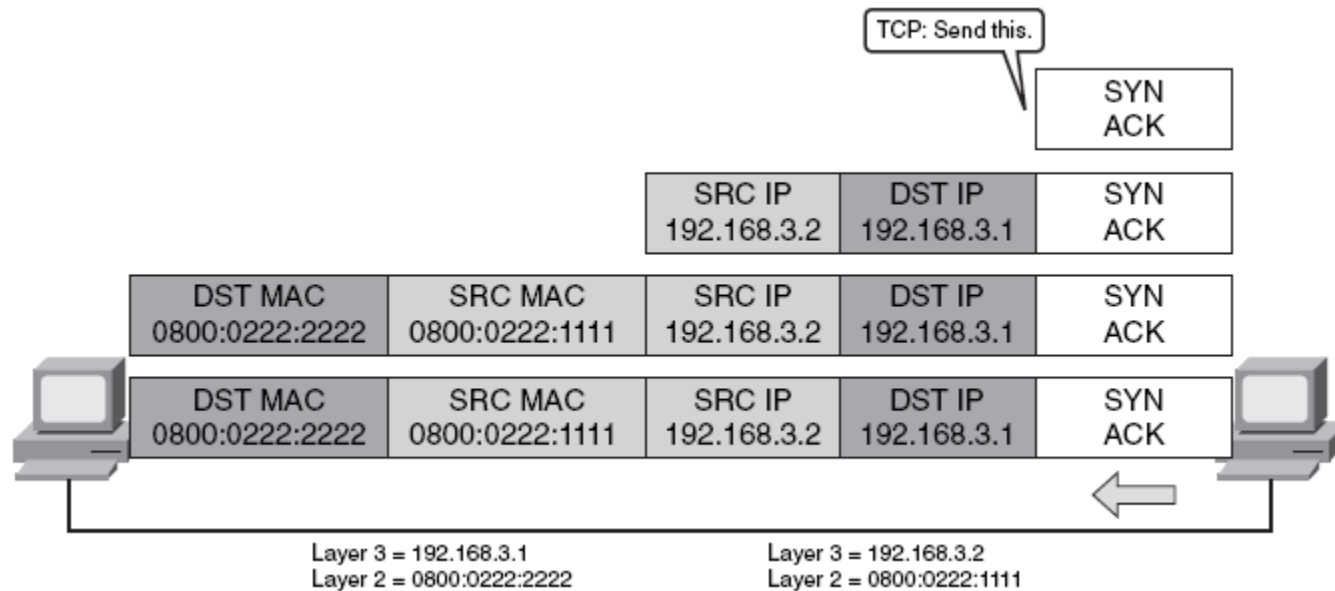


Sending SYN packet



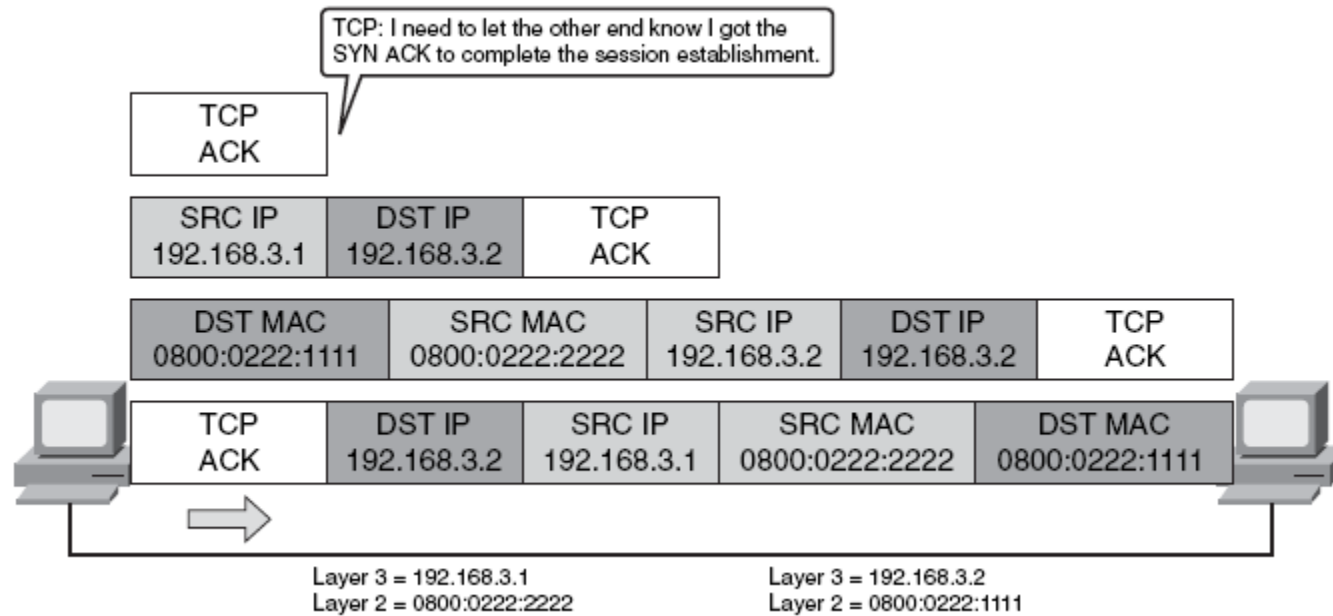


Reply SYN ACK



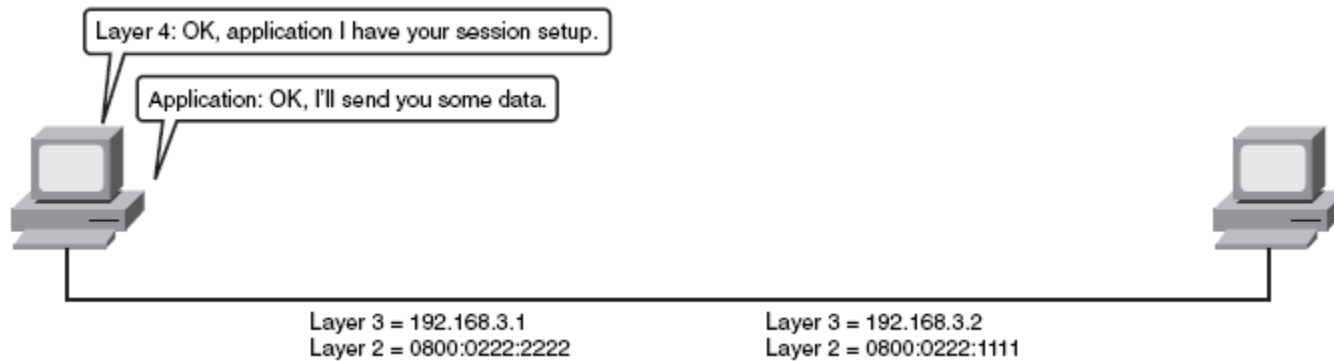


Send ACK



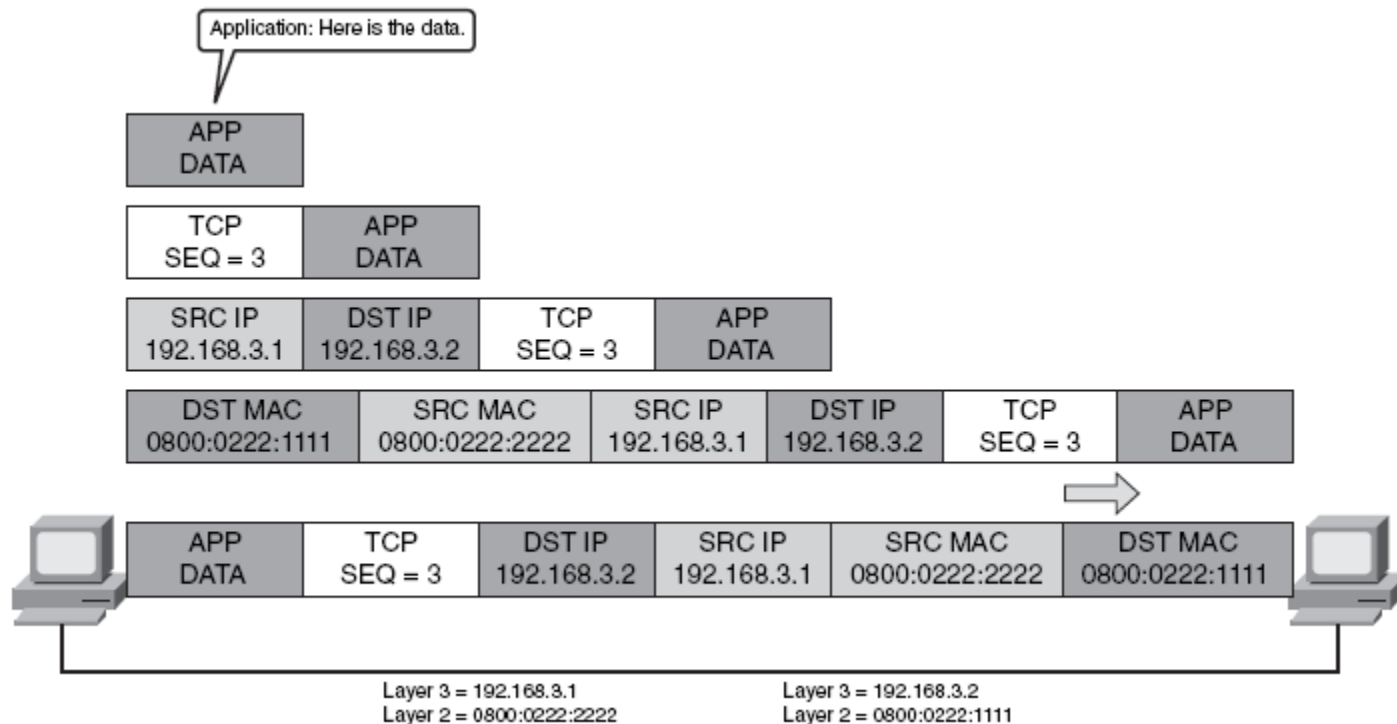


Open session





Data flow begins





Data is ACKed

